Human and mouse TNF-alpha binders

The selected 20: candidate histories and bound structures

Main text | 19 October 2026

Twenty candidates | Human and mouse TNF-alpha | Top and side structural views

We prioritize 20 exact protein sequences for human and mouse TNF-alpha, seeking interaction at pH 7.4 and release at pH 6.0. Each selected molecule descends from hotspot-conditioned Foundry/RFdiffusion3 generation followed by Proton-PottsMPNN sequence redesign. Seven are GS-linked two-domain fusions, four carry subsequent expert-specified substitutions, and nine retain their selected Proton-PottsMPNN sequence. We reconstruct the recorded steps for every candidate and show top and side views of all 120 selected human/mouse complexes predicted by Boltz2, OpenDDE, and AlphaFold3. Scores support computational prioritization.

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1 Introduction

The objective is a protein binder that recognizes soluble human TNF-alpha, cross-reacts with mouse TNF-alpha, and preferentially interacts at pH 7.4 rather than pH 6.0. We used complementary structural and sequence evidence to prioritize molecules for experimental assessment. The present main text is restricted to the 20 exact sequences in the October 19 ranked review CSV; its order is preserved throughout.

The initial strategy used AFSample3 conformational exploration of human and mouse TNF-alpha, including exploratory six-protomer systems, to guide hotspot choice. A target-only six-protomer input and a TNF trimer with three candidate copies are distinct molecular systems. The retained record supports the researcher-described role of conformational sampling in hotspot selection, but does not supply a complete sample-to-hotspot causal map for every candidate. The side and top epitope depictions are therefore retained as design context, without assigning an unrecorded sampled conformation to an individual molecule.

Hotspot-conditioned Foundry/RFdiffusion3 designs were refolded against both species and screened for interface confidence and structural preservation. Proton-PottsMPNN then proposed and optimized sequences on paired parent backbones for the two-pH objective. Selected descendants were either retained intact, changed by explicit expert substitutions, or combined as exact domains joined by geometry-guided GS linkers. Later three-copy predictions and copy-specific pH assessments tested the resulting exact sequences. Archived DeltaForge affinity predictions are reported separately with their earlier pose scope.

Each candidate section follows this sequence of evidence: original target and hotspot input; RFD3 source identity and sequence extraction; parent refolding and admission; Proton-PottsMPNN proposal and optimization; any subsequent mutation or fusion; downstream refolding and scoring; and current model-specific validation. Recorded settings and optimization trajectories support the candidate histories.

2 Shared methods and interpretation

2.1 Target constructs, hotspots, and source identity

The recorded paired constructs contain three human chains of 152 residues or three mouse chains of 148 residues, labeled A/B/C. Original source structures are 1TNF-associated human and 2TNF-associated mouse coordinates. Hotspots use source author numbering, not automatically the renumbered predictor input. The candidate histories name the exact side/top epitope cohort and author-numbered residues. Human soluble-domain reference and challenge sequence identities require separate review; the report preserves the actual modeled constructs rather than silently substituting another sequence.

Foundry/RFD3 source sequences were exported from native flat-CIF coordinate residue identities. The nine source scaffolds underlying the selected set have retained output specifications and design inputs. These provide the target contigs, fixed-target settings, hotspot atom selections, and source residue numbering used in each candidate history.

The human target contig is A6-157,/0,B6-157,/0,C6-157,/0,60-100. The mouse target contig is A9-72,A74-157,/0,B9-72,B74-157,/0,C9-72,C74-157,/0,60-100, retaining the coordinate-resolved gap at author residue 73. Both specify fixed target atoms, select_unfixed_sequence=false, and hotspot-based orientation, with one separate 60-100-residue de novo chain. Input parsing retains covalent/disulfide annotations and does not alter formal charges. Atom-level hotspot selectors appear in the candidate histories.

The side epitope is the initial hotspot set; the top epitope is the subsequently selected hotspot set. These names describe the chosen target surface regions. P and RP identify two parent-selection rounds, rather than the epitope itself. The source specifications record no fixed RFD3 seed; the recorded downstream redesign seeds are reported separately.

2.2 Parent selection and two-pH sequence redesign

Initial paired Boltz2 FAST assessments used 200 sampling steps, three recycles, one diffusion sample, and batch seed 0, with verified target MSAs and empty binder MSAs. The first parent selection retained ten from each species/epitope combination using mean human/mouse ipSAE multiplied by target-aligned human/mouse binder-pose TM similarity. A later top-epitope selection used mean ipSAE multiplied by mean RFD3-to-refolded binder-fold TM similarity. The labels P and RP identify these two parent-selection routes; they do not identify the source epitope by themselves. For example, P37 has a mouse top-epitope RFD3 ancestor.

Proton-PottsMPNN v6 used the epoch-0125 checkpoint. Conditional canonical decoder probabilities were averaged across the human and mouse parent backbones. Proposals were followed by 15 sweeps of two-pH Potts coordinate optimization. Temperatures were 0.05, 0.1, 0.2, and 0.4; each parent had a recorded seed. The pipeline built a pool of 160 accepted unique sequences to retain 100 per parent, subject to up to 2,000 attempts. Per-candidate proposal attempts, seeds, temperatures, mutable positions, exact substitutions, and objective values are reported below.

Up to 20 non-Cys binder positions nearest a target C-alpha within 10 angstrom in either species were mutable, ordered by summed distance. Target A/B/C, binder length, native Cys, and all other positions remained fixed, and new Cys were excluded. The generation-time interaction proxy was G = H(complex) - H(binder) - H(target trimer), with independent Henderson-Hasselbalch occupancies using intrinsic pKa H 6.5, D 3.9, and E 4.3. The minimized objective was 0.35 z(mean G7.4) + 0.40 z(mean G7.4 - mean G6.0) + 0.25 z(mean apo H7.4), with equal species weighting and frozen per-parent scales; z denotes division by the recorded scale, not recentering.

Acceptance required each species G7.4 to be no worse than its parent, G6.0 - G7.4 to be at least 0.10 model units and improve by at least 0.10 over the parent, apo H7.4 degradation no greater than 0.25 units per binder residue, and improvement of the composite objective. These are generation-time model guards. They neither calibrate affinity nor establish acidic release. Every retained exact descendant proceeded to paired human/mouse Boltz2 refolding. Optimization trajectories, baseline/scales, exported sequences and optimization records remain in each lineage record.

2.3 Expert substitutions and geometry-guided fusions

Expert variants apply only the named one-based substitutions to a selected Proton-PottsMPNN parent, leaving every other residue unchanged. They are deterministic sequence edits, followed by independent folding and scoring. They must not be described as an additional MPNN redesign. Candidate histories distinguish these expert edits from the preceding model-guided substitutions.

For fusions, exact two-domain sequences were concatenated in the geometry-selected order with a GGGGS-family linker. Parent-pair occupancy and receptor-overlay geometry preceded linker routing. A 2.0-angstrom grid with 3.0-angstrom atom-center exclusion sought solvent-accessible C-to-N paths. Linker length used 3.8 angstrom per residue at no more than 0.6 contour extension, at least five residues, a 3.0-angstrom separation between the three linker routes, and a 4.0-angstrom native-terminal launch zone. Candidate sections give the exact linker, domain boundaries, and maximum straight and routed distances. Parent overlays require receptor repacking and validation; favorable parent scores do not establish fusion performance.

The fusion generation record initially marked pH reassessment and refolding as pending. Subsequent Boltz2 Exact refolding and intact-fusion pH assessment supersede that generation-time status. Intact-fusion and extracted-domain observations remain distinct in the audit record. Three identical fusion chains D/E/F are modeled as separate copies; their two domains are covalently linked within each chain.

2.4 Current human/mouse models and pH assessment

The current evidence comprises one independently confidence-selected model for each species and method: Boltz2 Exact, OpenDDE Exact, and standard AlphaFold3. Each request generated five samples at seed 0; maximum native confidence selected the displayed sample, with the lower sample index resolving ties. Selection did not optimize pH or ipSAE. Boltz2 Exact used 200 sampling steps and three recycles; OpenDDE Exact used 200 diffusion steps and ten cycles; AlphaFold3 used ten recycles. Native target sequences, disulfides, chain roles, input preparation, cached MSA/features, and method-specific software pins are retained in the source study. Standard AlphaFold3 here is separate from initial AFSample3 exploration.

Each model contains target A/B/C and three separate identical candidate copies D/E/F. Copy-specific ipSAE uses native PAE, a 10-angstrom cutoff, and best-protomer scoring; the weakest candidate copy is reported. The review threshold is 0.50. Cross-species agreement and each of three method pairs per species use symmetry-aware assembly DockQ, requiring at least 0.49 aggregate and 0.23 for the worst matched copy. These yield three cross-species method comparisons and six intermethod comparisons per candidate. Confidence and geometric agreement are computational diagnostics.

Post-refolding Proton-PottsMPNN assessment uses the selected native poses, selected binder His/Asp/Glu centers, and bound-complex PROPKA estimates. The same sequence positions are retained across species, methods, and copies, including translated domain positions in fusions. Each D/E/F copy is assessed against A/B/C separately, with receptor and noncenter ionization labels fixed. Independent-site expectations are evaluated at pH 6.0 and 7.4 under common pKa shifts -1, 0, and +1. The contrast S6.0 - S7.4 is positive when the proxy favors pH 7.4.

Potts interaction contrasts are in native model units; canonical teacher-forced MPNN contrasts are in nats per residue. Each model table reports the minimum over all copies and all pKa scenarios separately for both proxies. A robust-positive model requires both minima to be positive. The count out of six is evidence coverage across species/method combinations, not independent biological replication. Isolated-copy scoring does not estimate trivalent avidity, coupled protonation, or the full assembly binding equilibrium.

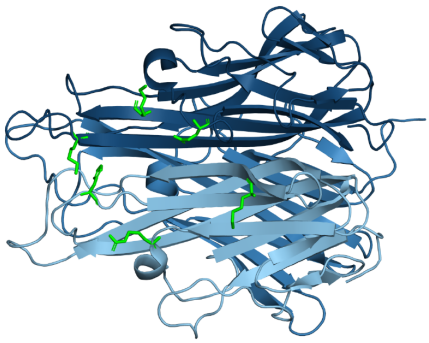
2.5 Archived affinity and selection scope

All 20 sequences have saved human and mouse DeltaForge predicted KD values. The raw score receipts identify target A/B/C plus a single candidate D: either an earlier single-binder refold or one intact fusion chain extracted from an earlier A/B/C+D/E/F model with E/F omitted. No new affinity prediction was performed for the displayed six-chain ensembles. Values are native nM, without a pH input. They are neither measured affinities nor current assembly KD values. Raw DeltaG is retained in the audit evidence in its native units, without an unsupported conversion.

The source selection retains four explicit manual overrides: P37-d002, F01, RP03-d017, and P37-d022. For other entries, at least one of the six selected models must have positive Potts and MPNN contrasts across all copies and pKa scenarios. Ordering prioritizes robust-positive model count, intermethod agreement count, cross-species agreement count, lower worse-species archived KD, ipSAE coverage, Boltz2 minimum-copy ipSAE, and stable ID. We preserve the supplied CSV order and perform no new ranking. All entries remain research-review candidates; the ranked set is defined, but validation flags and manual inclusion are not altered by its membership.

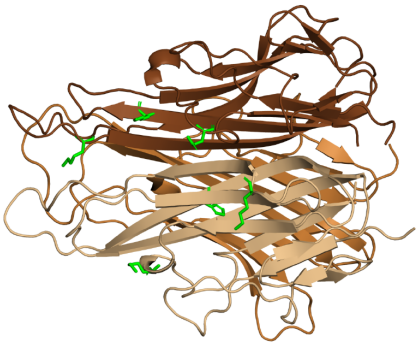
3 Side and top epitope context

A Human TNF-alpha - side epitope



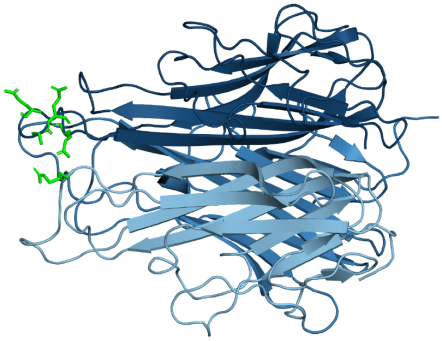
A: K65, K112, E146; C: H73, K90, R138

B Mouse TNF-alpha - side epitope



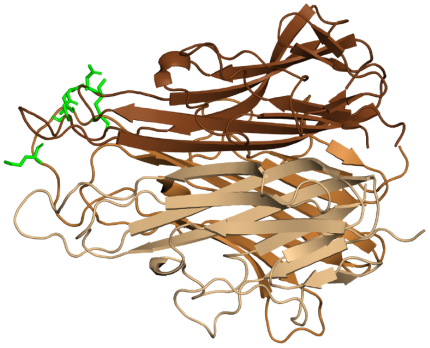
A: K65, K112, E146; C: H78, K90, K140

C Human TNF-alpha - top epitope



A: E104, E107, A109, E110; C: R103

D Mouse TNF-alpha - top epitope



A: D71, D104; B: E107, E110, K112

Green sticks: hotspot atoms | Target shades: protomers A/B/C | Source author residue numbering

Figure 1. Side and top epitopes on human and mouse TNF-alpha. A, human side epitope: A K65/K112/E146; C H73/K90/R138. B, mouse side epitope: A K65/K112/E146; C H78/K90/K140. C, human top epitope: A E104/E107/A109/E110; C R103. D, mouse top epitope: A D71/D104; B E107/E110/K112. Residues use source author numbering. Three blue shades identify human target protomers, three copper shades identify mouse target protomers, and green marks hotspot atoms. All four structural depictions retain their original common orientation and scale.

The retained structural panels depict the selected epitopes. A complete AFSample3 sample-to-epitope selection and clustering panel remains unavailable.

4 The selected 20

Table 1. Exact CSV order. pH is the number of robust-positive selected species/method models out of six; pose is passing intermethod pairs out of six. KD values are archived human/mouse single-binder predictions in nM. An asterisk marks a manual override. Each entry has a full section and paired six-model top and side view figures below.

Rank	Candidate	Sequence class	pH /6	Pose /6	KD H / M (nM)
1	P37-d002*	Proton-Potts redesign	6	6	42.48 / 43.05
2	RP06-d085-V60T	expert mutant	6	6	64.01 / 63.35
3	F07	GS fusion	6	1	21.19 / 15.17
4	F01*	GS fusion	6	1	37.71 / 13.22
5	RP03-d017*	Proton-Potts redesign	5	4	28.63 / 44.82
6	P18-d092	Proton-Potts redesign	5	4	48.28 / 33.26
7	P12-d084	Proton-Potts redesign	5	3	57.31 / 54.44
8	F02	GS fusion	5	1	15.85 / 15.41
9	P37-d022*	Proton-Potts redesign	4	6	50.84 / 44.36
10	RP06-d085	Proton-Potts redesign	4	4	42.21 / 46.40
11	F09	GS fusion	4	2	21.12 / 17.21
12	P02-d070	Proton-Potts redesign	4	1	49.88 / 59.55
13	F25	GS fusion	3	4	45.58 / 33.23
14	RP06-d085-V60S	expert mutant	3	2	65.39 / 65.44
15	F11	GS fusion	3	0	37.18 / 21.21
16	F32	GS fusion	2	6	37.37 / 23.37
17	RP03-d050-A40N_A69Q	expert mutant	2	4	62.17 / 73.52
18	RP03-d043-A40N_A69Q	expert mutant	2	4	79.22 / 74.35
19	P16-d085	Proton-Potts redesign	1	6	29.04 / 24.05
20	P00-d000	Proton-Potts redesign	1	1	17.26 / 13.07

The set comprises nine Proton-PottsMPNN redesigns, four subsequent expert mutants, and seven two-domain GS fusions. Candidate sections use one-based positions within each isolated parent sequence; fusion domain boundaries then map these positions onto the complete submitted chain. Only the 20 selected exact sequences are reported as candidates here; necessary ancestral domains are named as provenance intermediates.

5.01 P37-d002

Rank 1 | Proton-Potts redesign | 60 residues | Manual override retained

Candidate: P37-d002. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P37-d002: creation history

RFD3 source scaffold. The source scaffold for P37-d002 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d002. The recorded redesign seed is 3700111, accepted proposal attempt 2, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -412.893665; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V. The 11 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 12.9270, G6.0 - G7.4 = 4.7342; mouse G7.4 = -25.1116, G6.0 - G7.4 = 2.6896. G7.4 changes versus the parent are -7.8463 and -4.0544, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 42.48 nM; mouse 43.05 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

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EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRV
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5.01 P37-d002 - bound structures

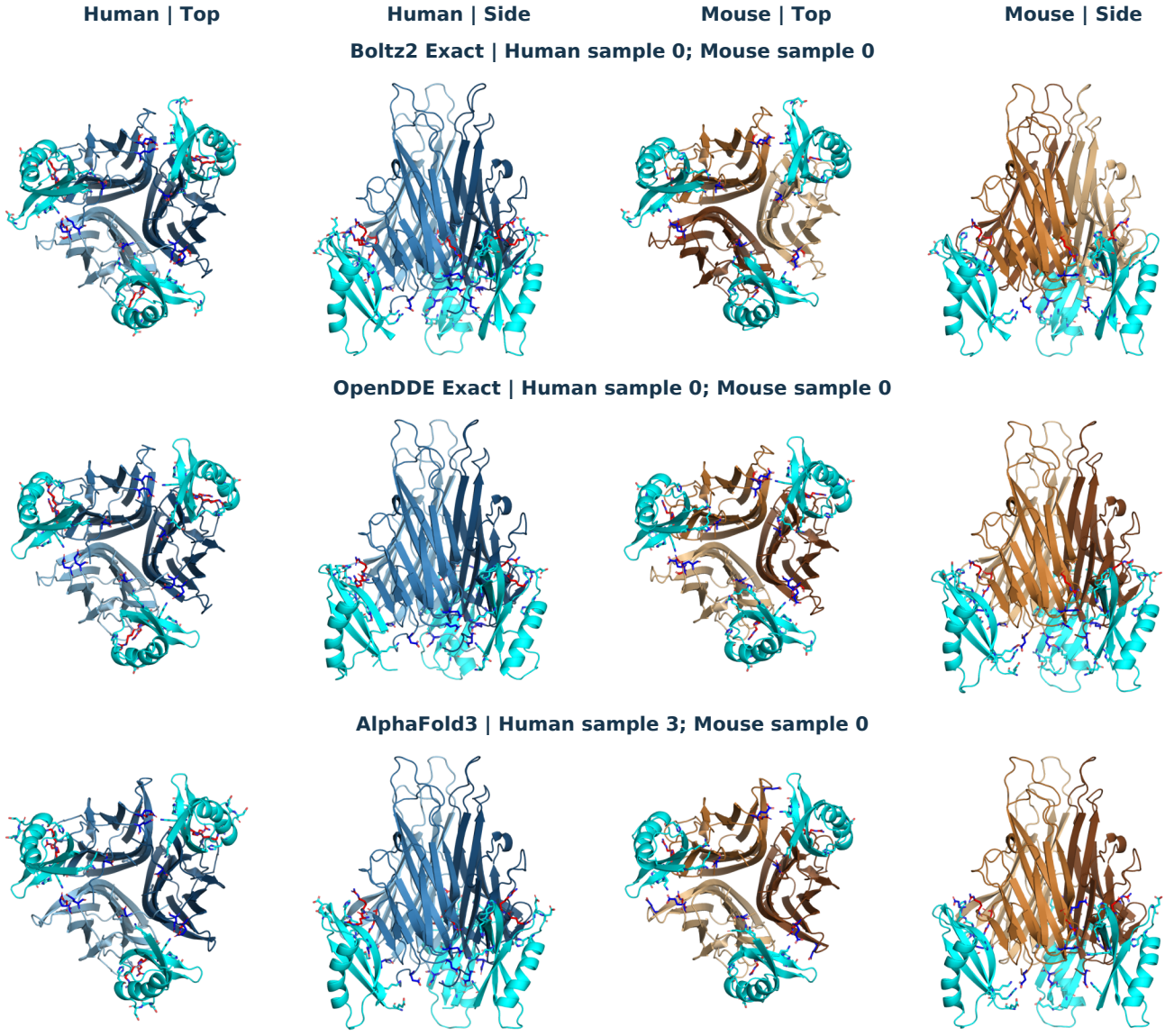

Figure 2. P37-d002: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.846	0.1679	0.005992
Boltz2 / mouse / 0	0.834	0.02114	0.002118
OpenDDE / human / 0	0.040	0.9324	0.001971
OpenDDE / mouse / 0	0.210	0.105	0.003305
AF3 / human / 3	0.463	0.1033	0.004131
AF3 / mouse / 0	0.000	0.04628	0.004848

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45.

5.02 RP06-d085-V60T

Rank 2 | expert mutant | 83 residues | Evidence-tier retained

Candidate: RP06-d085-V60T. The structures and metrics below refer to its exact selected sequence.

Creation route. Foundry/RFD3 generation and Proton-PottsMPNN redesign produced RP06-d085; the final sequence then received expert-specified V60T substitution(s). The model-guided and expert-specified edits are recorded separately below.

RP06-d085: creation history

RFD3 source scaffold. The source scaffold for RP06-d085 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 83-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 7; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP06-d085. The recorded redesign seed is 20861028, accepted proposal attempt 87, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -715.992329 in the parent to -719.792428; these are model objective values.

Mutable one-based positions. 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T. The 13 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 144.7273, G6.0 - G7.4 = 3.6622; mouse G7.4 = 46.6635, G6.0 - G7.4 = 1.5454. G7.4 changes versus the parent are -10.7404 and -0.2608, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact expert mutation step

Expert-specified V60T substitutions were applied to RP06-d085. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP06-d085-V60T. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 64.01 nM; mouse 63.35 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGTHTYTCTATGA
GGTNGASTTVTVT
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5.02 RP06-d085-V60T - bound structures

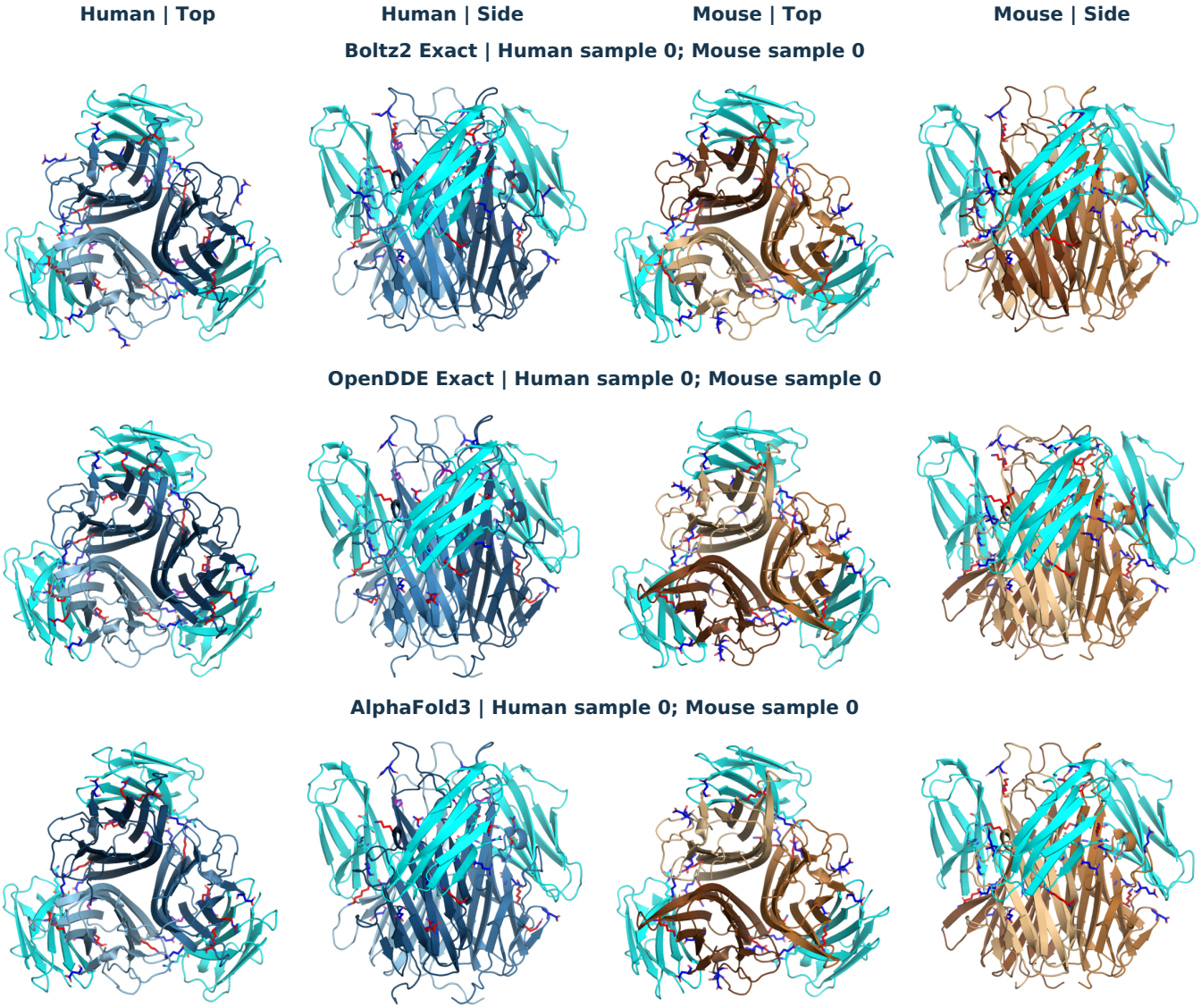


Figure 3. RP06-d085-V60T: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 6/6; ipSAE pass: 4/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.845	0.004502	1.939e-05
Boltz2 / mouse / 0	0.821	0.009326	5.188e-05
OpenDDE / human / 0	0.543	2.341	0.01703
OpenDDE / mouse / 0	0.577	0.004258	8.017e-05
AF3 / human / 0	0.000	0.002634	2.694e-05
AF3 / mouse / 0	0.000	0.008006	6.372e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.03 F07

Rank 3 | GS fusion | 163 residues | Evidence-tier retained

Candidate: F07. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P37-d022 and RP03-d050, were joined in that N-to-C order through an exact 22-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P37-d022: creation history

RFD3 source scaffold. The source scaffold for P37-d022 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d022. The recorded redesign seed is 3700111, accepted proposal attempt 22, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -413.550655; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 13.3211, G6.0 - G7.4 = 5.7922; mouse G7.4 = -23.4156, G6.0 - G7.4 = 2.5233. G7.4 changes versus the parent are -7.4522 and -2.3585, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.03 F07 - second domain history

Domain 2 - RP03-d050: creation history

RFD3 source scaffold. The source scaffold for RP03-d050 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d050. The recorded redesign seed is 20561019, accepted proposal attempt 73, and temperature 0.1, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -670.558797; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T21S, S28T, A49H, S51H, Q67T. The 5 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 54.8035, G6.0 - G7.4 = 2.4292; mouse G7.4 = 115.1998, G6.0 - G7.4 = 0.4890. G7.4 changes versus the parent are -10.0744 and -4.9792, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d050. The maximum C-to-N distance over the paired geometry was 47.98 angstrom and maximum solvent-route length 50.88 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 21.19 nM; mouse 15.17 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

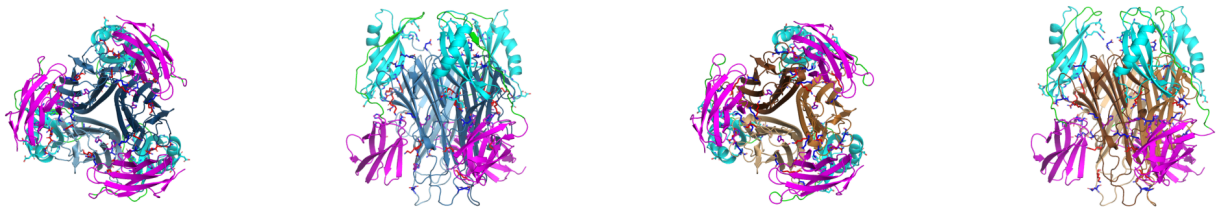
Exact selected sequence (N to C):

```
EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGATIAAAAATHAHFTVPATP
GTYTVTAATGADASSATVTVAAA
```

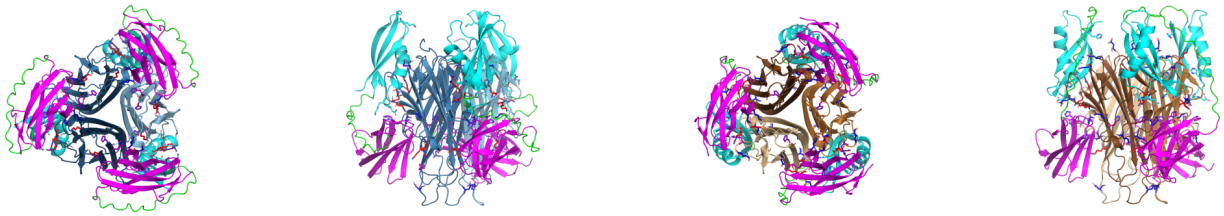
5.03 F07 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 0; Mouse sample 1

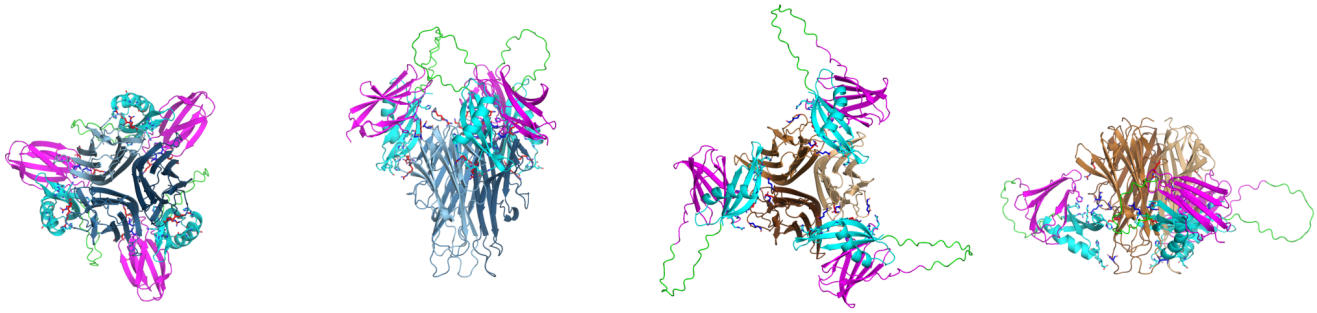


Figure 4. F07: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.744	0.0881	0.0007851
Boltz2 / mouse / 0	0.744	0.1232	0.001698
OpenDDE / human / 0	0.074	1.012	0.0011
OpenDDE / mouse / 0	0.039	0.07099	0.001089
AF3 / human / 0	0.000	0.6119	0.003354
AF3 / mouse / 1	0.000	0.0172	0.0004073

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 133, 131.

5.04 F01

Rank 4 | GS fusion | 165 residues | Manual override retained

Candidate: F01. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P37-d002 and RP03-d017, were joined in that N-to-C order through an exact 24-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P37-d002: creation history

RFD3 source scaffold. The source scaffold for P37-d002 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d002. The recorded redesign seed is 3700111, accepted proposal attempt 2, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -412.893665; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V. The 11 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 12.9270, G6.0 - G7.4 = 4.7342; mouse G7.4 = -25.1116, G6.0 - G7.4 = 2.6896. G7.4 changes versus the parent are -7.8463 and -4.0544, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.04 F01 - second domain history

Domain 2 - RP03-d017: creation history

RFD3 source scaffold. The source scaffold for RP03-d017 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d017. The recorded redesign seed is 20561019, accepted proposal attempt 23, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -672.193798; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N. The 8 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 53.9802, G6.0 - G7.4 = 0.7972; mouse G7.4 = 108.5008, G6.0 - G7.4 = 3.3428. G7.4 changes versus the parent are -10.8977 and -11.6783, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d002 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 52.65 angstrom and maximum solvent-route length 55.33 angstrom. Domain 1 occupies residues 1-60, the linker 61-84, and domain 2 85-165. Exact linker: GGGGSGGGGSGGGGSGGGGSGGGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 37.71 nM; mouse 13.22 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

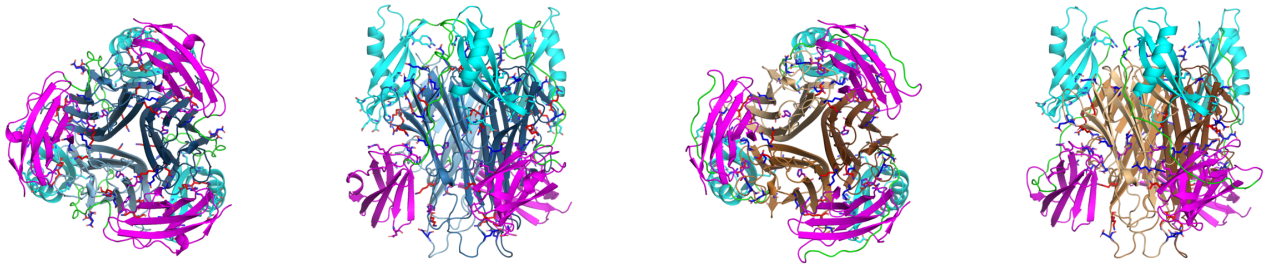
Exact selected sequence (N to C):

```
EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATIAAAAAARATFTVPA
TPGTYTVTAANGADASSATVTVAAA
```

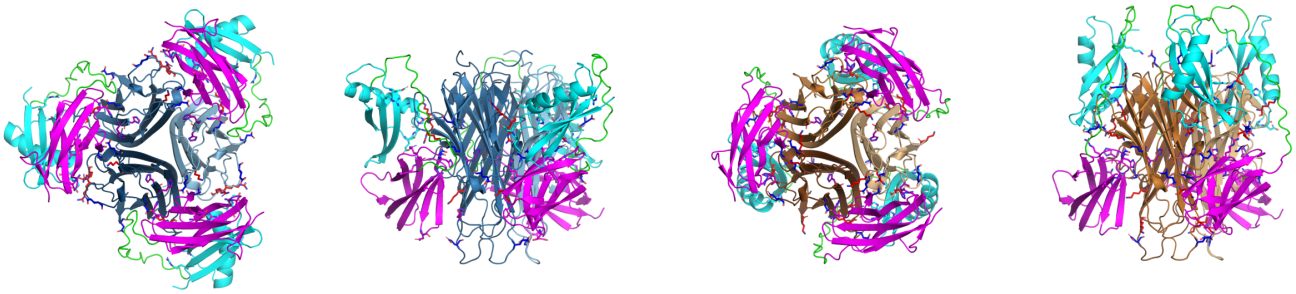
5.04 F01 - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 1; Mouse sample 3

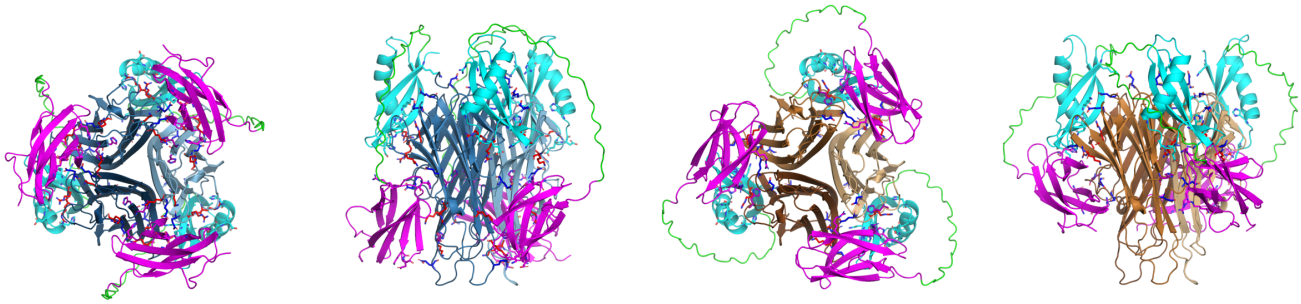


Figure 5. F01: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 6/6; ipSAE pass: 2/6; cross-species pose pass: 0/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.710	0.04959	0.0009204
Boltz2 / mouse / 0	0.771	0.03726	0.0004203
OpenDDE / human / 0	0.047	0.1587	0.006539
OpenDDE / mouse / 0	0.037	0.2312	0.0005135
AF3 / human / 1	0.000	0.2142	0.001249
AF3 / mouse / 3	0.000	0.001968	7.6e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45, 114, 122.

5.05 RP03-d017

Rank 5 | Proton-Potts redesign | 81 residues | Manual override retained

Candidate: RP03-d017. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

RP03-d017: creation history

RFD3 source scaffold. The source scaffold for RP03-d017 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d017. The recorded redesign seed is 20561019, accepted proposal attempt 23, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -672.193798; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N. The 8 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 53.9802, G6.0 - G7.4 = 0.7972; mouse G7.4 = 108.5008, G6.0 - G7.4 = 3.3428. G7.4 changes versus the parent are -10.8977 and -11.6783, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 28.63 nM; mouse 44.82 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
AAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATIAAAAAARATFTVPATPGTYTVTAANGAD
ASSATVTVAAA
```

5.05 RP03-d017 - bound structures

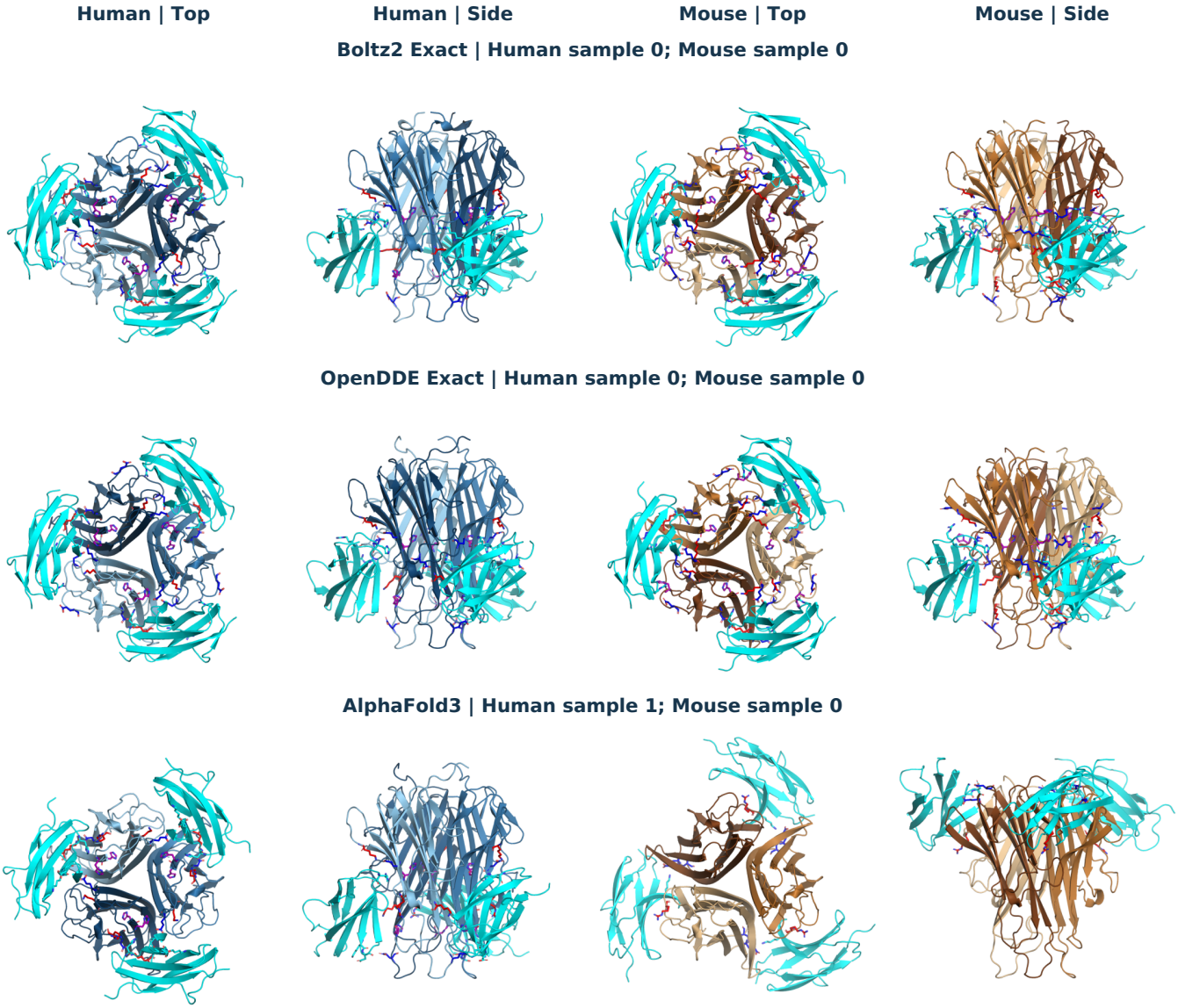


Figure 6. RP03-d017: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.824	0.002432	2.566e-05
Boltz2 / mouse / 0	0.843	0.002364	5.503e-05
OpenDDE / human / 0	0.449	0.01061	3.598e-05
OpenDDE / mouse / 0	0.387	0.01301	3.499e-05
AF3 / human / 1	0.202	0.003532	6.983e-05
AF3 / mouse / 0	0.000	-0.4338	-0.004906

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 30, 38.

5.06 P18-d092

Rank 6 | Proton-Potts redesign | 73 residues | Evidence-tier retained

Candidate: P18-d092. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P18-d092: creation history

RFD3 source scaffold. The source scaffold for P18-d092 used side epitope / mouse conditioning on the complete target trimer: A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 73-residue parent.

Exact hotspot atom selectors. A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 9; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P18-d092. The recorded redesign seed is 1800054, accepted proposal attempt 94, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -575.859055 in the parent to -582.311822; these are model objective values.

Mutable one-based positions. 20, 21, 22, 23, 24, 25, 26, 27, 28, 44, 45, 54, 55, 56, 57, 58, 59, 60, 62, 63. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. R23A, S24P, A27K, T28V, D44T, A45N, E55Q, T56H, F57Y, I58H, A60G, K62R. The 12 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 3.1359, G6.0 - G7.4 = 3.0980; mouse G7.4 = -9.1645, G6.0 - G7.4 = 2.4289. G7.4 changes versus the parent are -20.6686 and -17.7771, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 48.28 nM; mouse 33.26 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
MLSASELQAGDIVSLPSSVKAIAPGGKVIDLPAGSYKVIAVSSTNIALLDAATGQHYHVGSRVLNQNGGV
TLM
```

5.06 P18-d092 - bound structures

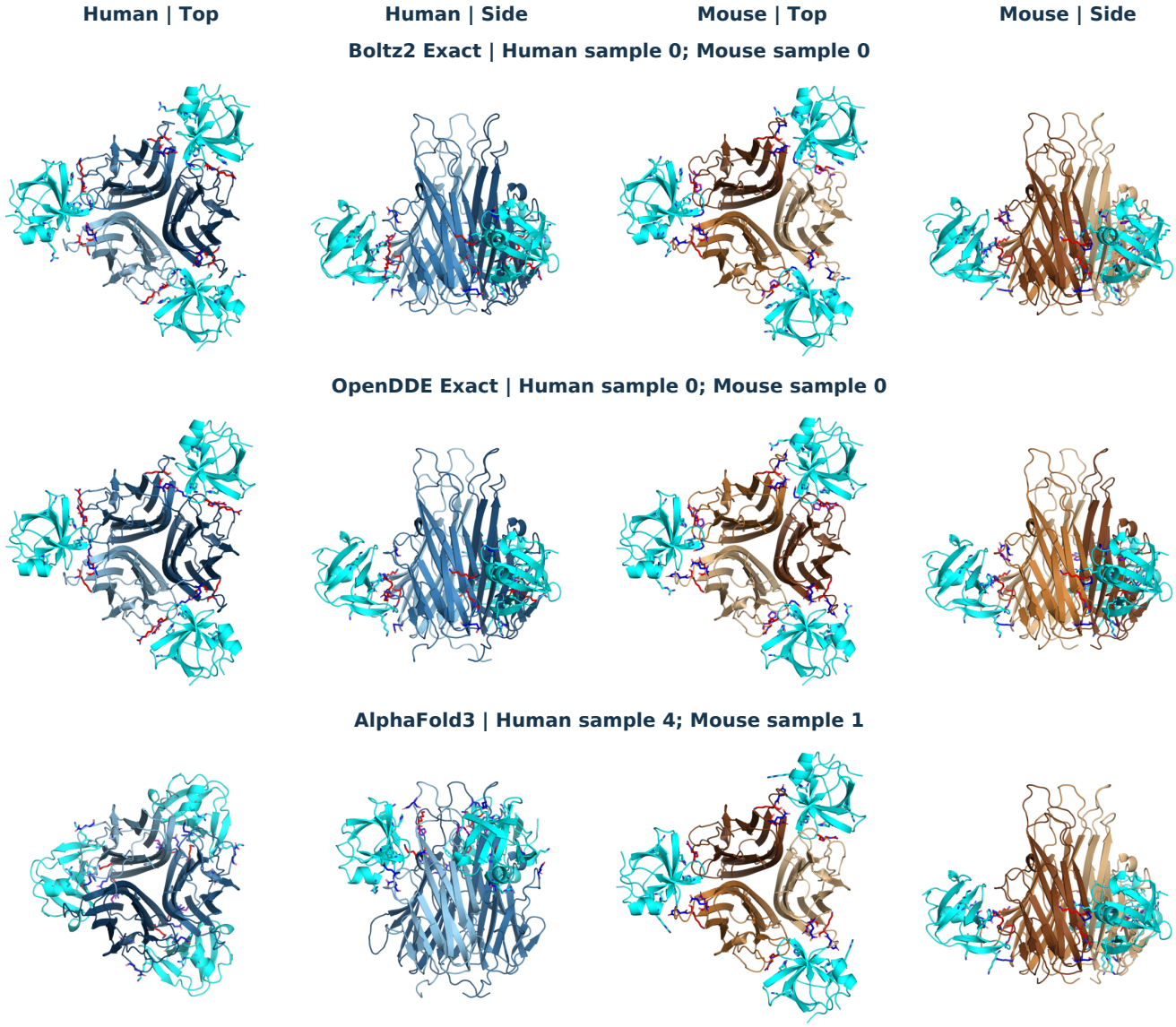


Figure 7. P18-d092: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 4/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.708	0.008808	0.000252
Boltz2 / mouse / 0	0.651	0.0119	-2.193e-05
OpenDDE / human / 0	0.547	0.00437	5.904e-05
OpenDDE / mouse / 0	0.518	0.002171	4.89e-05
AF3 / human / 4	0.000	0.0112	0.0005285
AF3 / mouse / 1	0.464	0.002491	8.589e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 58, 56.

5.07 P12-d084

Rank 7 | Proton-Potts redesign | 75 residues | Evidence-tier retained

Candidate: P12-d084. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P12-d084: creation history

RFD3 source scaffold. The source scaffold for P12-d084 used side epitope / mouse conditioning on the complete target trimer: A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 75-residue parent.

Exact hotspot atom selectors. A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 3; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P12-d084. The recorded redesign seed is 1200036, accepted proposal attempt 90, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -560.353410 in the parent to -563.850861; these are model objective values.

Mutable one-based positions. 22, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 59, 60, 61, 62. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. Q22N, G35V, A36R, A42S, G43H, N48D, I49V, T59L, V61R, A62K. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 72.0461, G6.0 - G7.4 = 0.6854; mouse G7.4 = 52.5755, G6.0 - G7.4 = 1.0881. G7.4 changes versus the parent are -13.9978 and -12.1394, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 57.31 nM; mouse 54.44 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
MKIEISVKNGIAQVAVTAVSENDVYTITLAGFNVVRVLAVSSHLTVLDVNAATGTLTYLARKGQDGVIVL
AVVPR
```

5.07 P12-d084 - bound structures

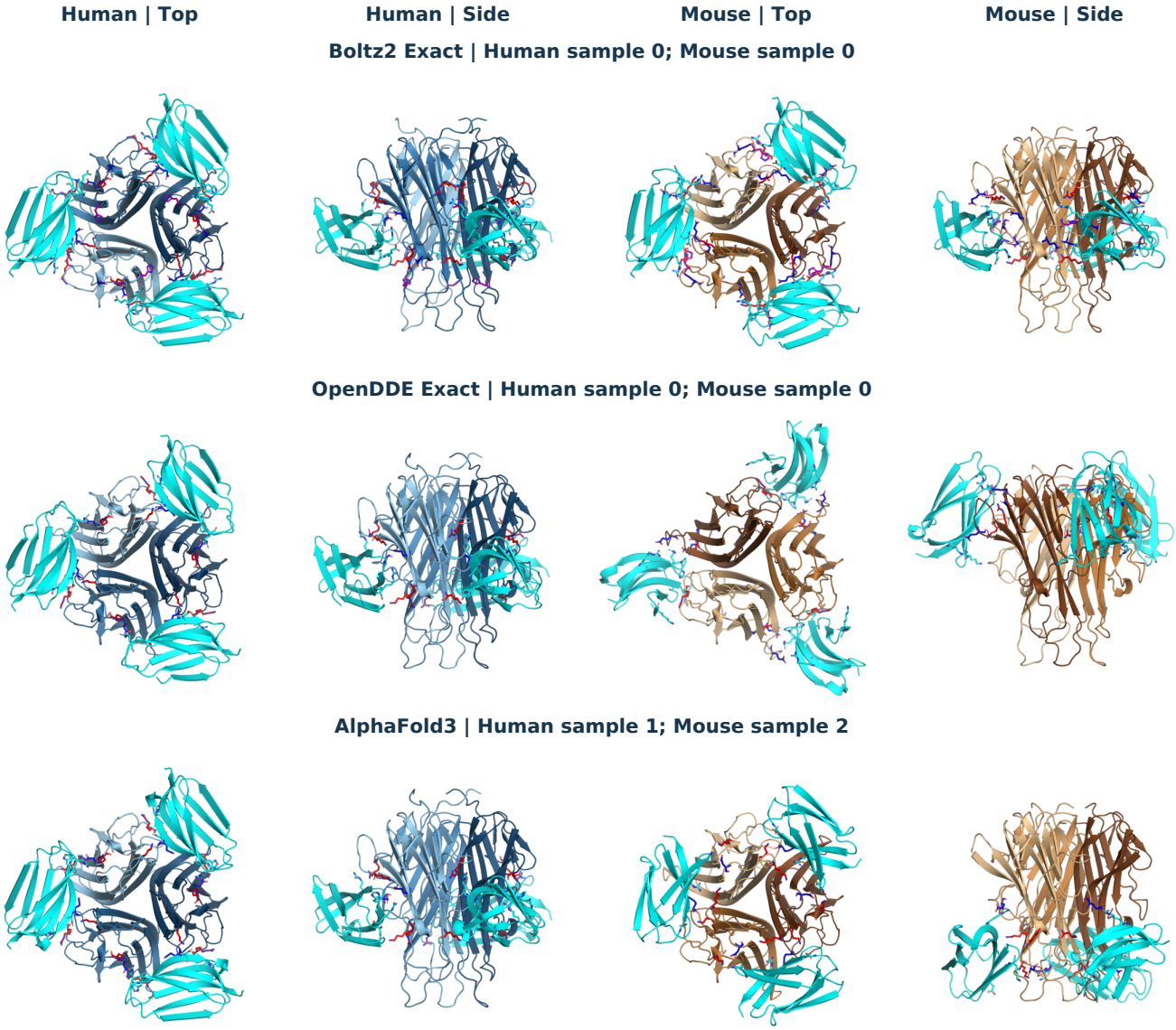


Figure 8. P12-d084: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 5/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 3/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.832	-1.248	0.01024
Boltz2 / mouse / 0	0.809	0.007753	5.776e-05
OpenDDE / human / 0	0.308	0.02529	0.0005514
OpenDDE / mouse / 0	0.018	0.009171	0.0005305
AF3 / human / 1	0.526	0.02808	0.000416
AF3 / mouse / 2	0.000	0.006195	0.0004766

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 48, 21.

5.08 F02

Rank 8 | GS fusion | 163 residues | Evidence-tier retained

Candidate: F02. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P37-d002 and RP03-d050, were joined in that N-to-C order through an exact 22-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P37-d002: creation history

RFD3 source scaffold. The source scaffold for P37-d002 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d002. The recorded redesign seed is 3700111, accepted proposal attempt 2, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -412.893665; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3K, A12E, A16H, A39P, V40I, I42V, V45H, Q46H, T47A, Q48P, I53V. The 11 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 12.9270, G6.0 - G7.4 = 4.7342; mouse G7.4 = -25.1116, G6.0 - G7.4 = 2.6896. G7.4 changes versus the parent are -7.8463 and -4.0544, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.08 F02 - second domain history

Domain 2 - RP03-d050: creation history

RFD3 source scaffold. The source scaffold for RP03-d050 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d050. The recorded redesign seed is 20561019, accepted proposal attempt 73, and temperature 0.1, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -670.558797; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T21S, S28T, A49H, S51H, Q67T. The 5 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 54.8035, G6.0 - G7.4 = 2.4292; mouse G7.4 = 115.1998, G6.0 - G7.4 = 0.4890. G7.4 changes versus the parent are -10.0744 and -4.9792, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d002 → linker → RP03-d050. The maximum C-to-N distance over the paired geometry was 47.86 angstrom and maximum solvent-route length 50.30 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 15.85 nM; mouse 15.41 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
EAKVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPIAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGATIAAAAATHAHFTVPATP
GTYTVTAATGADASSATVTVAAA
```

5.08 F02 - bound structures

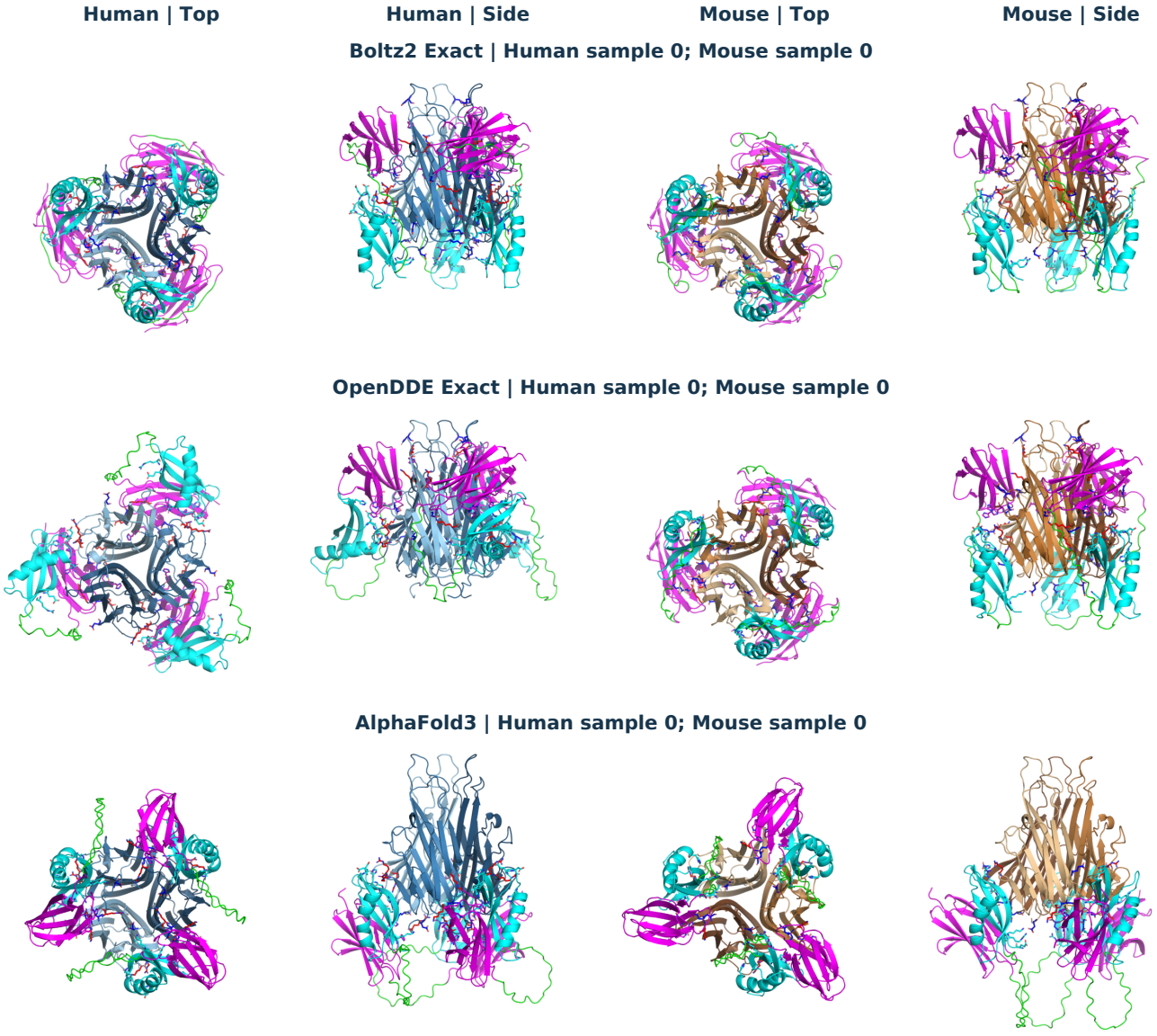


Figure 9. F02: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 5/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.662	0.04616	0.0011
Boltz2 / mouse / 0	0.720	0.7341	0.0125
OpenDDE / human / 0	0.067	0.3992	0.016
OpenDDE / mouse / 0	0.038	0.08544	0.002243
AF3 / human / 0	0.000	0.104	0.0004331
AF3 / mouse / 0	0.000	-0.6358	0.01128

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 45, 133, 131.

5.09 P37-d022

Rank 9 | Proton-Potts redesign | 60 residues | Manual override retained

Candidate: P37-d022. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P37-d022: creation history

RFD3 source scaffold. The source scaffold for P37-d022 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d022. The recorded redesign seed is 3700111, accepted proposal attempt 22, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -413.550655; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 13.3211, G6.0 - G7.4 = 5.7922; mouse G7.4 = -23.4156, G6.0 - G7.4 = 2.5233. G7.4 changes versus the parent are -7.4522 and -2.3585, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 50.84 nM; mouse 44.36 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRV
```

5.09 P37-d022 - bound structures

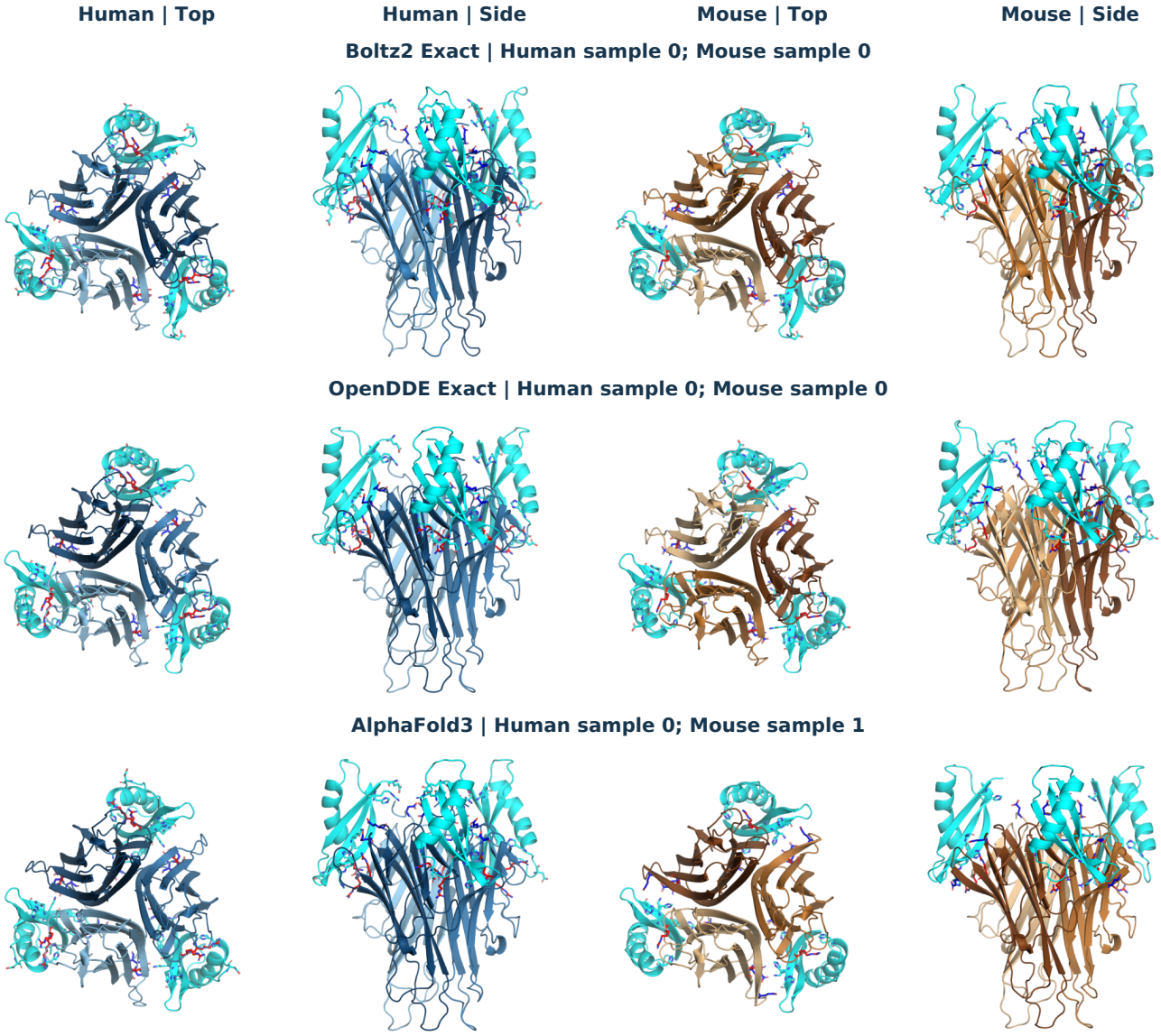


Figure 10. P37-d022: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.878	0.05376	0.001811
Boltz2 / mouse / 0	0.769	0.0455	0.001771
OpenDDE / human / 0	0.178	0.8033	-0.01301
OpenDDE / mouse / 0	0.034	0.2684	0.008911
AF3 / human / 0	0.435	0.0928	0.003157
AF3 / mouse / 1	0.000	-0.118	0.008269

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1.

5.10 RP06-d085

Rank 10 | Proton-Potts redesign | 83 residues | Evidence-tier retained

Candidate: RP06-d085. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

RP06-d085: creation history

RFD3 source scaffold. The source scaffold for RP06-d085 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 83-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 7; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP06-d085. The recorded redesign seed is 20861028, accepted proposal attempt 87, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -715.992329 in the parent to -719.792428; these are model objective values.

Mutable one-based positions. 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T. The 13 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 144.7273, G6.0 - G7.4 = 3.6622; mouse G7.4 = 46.6635, G6.0 - G7.4 = 1.5454. G7.4 changes versus the parent are -10.7404 and -0.2608, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 42.21 nM; mouse 46.40 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGVHTYTCTATGA
GGTNGASTTVTVT
```

5.10 RP06-d085 - bound structures

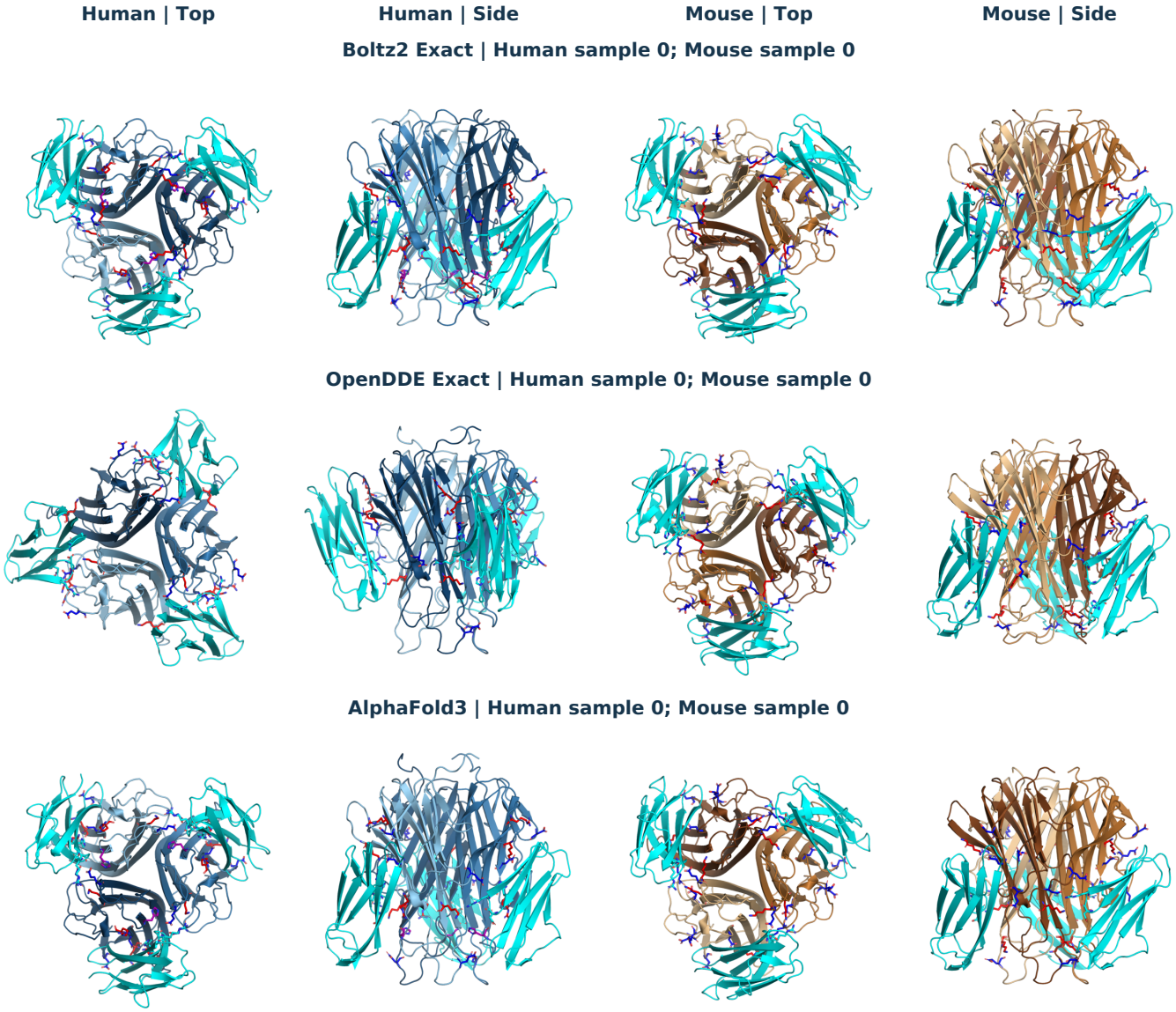


Figure 11. RP06-d085: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 4/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.825	0.004672	2.269e-05
Boltz2 / mouse / 0	0.855	0.01145	0.0001187
OpenDDE / human / 0	0.016	0.05624	-0.005319
OpenDDE / mouse / 0	0.585	0.006379	1.988e-05
AF3 / human / 0	0.619	-0.0006657	-0.0005708
AF3 / mouse / 0	0.000	0.00963	7e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.11 F09

Rank 11 | GS fusion | 164 residues | Evidence-tier retained

Candidate: F09. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P37-d022 and RP03-d043, were joined in that N-to-C order through an exact 23-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P37-d022: creation history

RFD3 source scaffold. The source scaffold for P37-d022 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d022. The recorded redesign seed is 3700111, accepted proposal attempt 22, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -413.550655; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 13.3211, G6.0 - G7.4 = 5.7922; mouse G7.4 = -23.4156, G6.0 - G7.4 = 2.5233. G7.4 changes versus the parent are -7.4522 and -2.3585, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.11 F09 - second domain history

Domain 2 - RP03-d043: creation history

RFD3 source scaffold. The source scaffold for RP03-d043 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d043. The recorded redesign seed is 20561019, accepted proposal attempt 64, and temperature 0.05, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -669.254062; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, A27H, T32L, T48S, A49R, S51T. The 6 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 57.6741, G6.0 - G7.4 = 0.8766; mouse G7.4 = 104.6333, G6.0 - G7.4 = 0.1625. G7.4 changes versus the parent are -7.2038 and -15.5458, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d043. The maximum C-to-N distance over the paired geometry was 50.61 angstrom and maximum solvent-route length 53.08 angstrom. Domain 1 occupies residues 1-60, the linker 61-83, and domain 2 84-164. Exact linker: GGGGSGGGGSGGGGSGGGGSGGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 21.12 nM; mouse 17.21 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGGSAAGPAVSAGASTTVTAGARLTVSASNHSGAVLTTLSKDGATIAAAAASRATFTVPAT
PGTYTVTAAQGADASSATVTVAAA
```

5.11 F09 - bound structures

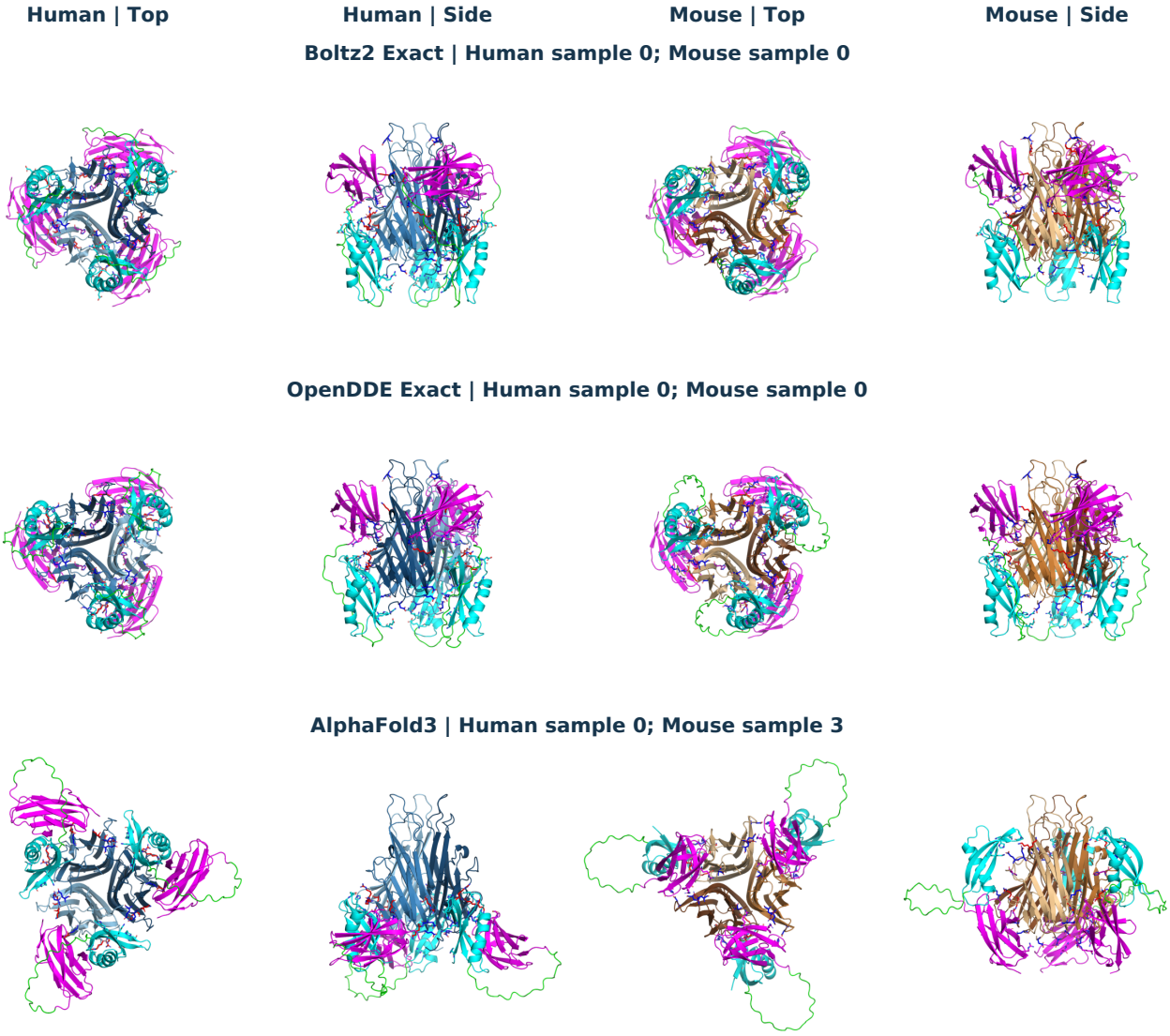


Figure 12. F09: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 4/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 2/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.764	0.04201	0.0002242
Boltz2 / mouse / 0	0.757	0.04487	0.001143
OpenDDE / human / 0	0.033	0.9914	0.002518
OpenDDE / mouse / 0	0.043	0.7582	-0.01249
AF3 / human / 0	0.000	0.04124	0.005063
AF3 / mouse / 3	0.000	-0.02139	-0.00675

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 110, 153.

5.12 P02-d070

Rank 12 | Proton-Potts redesign | 73 residues | Evidence-tier retained

Candidate: P02-d070. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P02-d070: creation history

RFD3 source scaffold. The source scaffold for P02-d070 used side epitope / human conditioning on the complete target trimer: A:LYS65, A:LYS112, A:GLU146, C:HIS73, C:LYS90, C:ARG138. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 73-residue parent.

Exact hotspot atom selectors. A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C73: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C138: CB,CG,CD,NE,CZ,NH1,NH2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 3; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P02-d070. The recorded redesign seed is 200006, accepted proposal attempt 74, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -550.317242 in the parent to -558.700326; these are model objective values.

Mutable one-based positions. 15, 16, 17, 19, 24, 25, 26, 27, 31, 32, 33, 58, 60, 61, 62, 63, 64, 65, 66, 67. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. Q19S, N25Q, A31P, L33R, N58H, I60W, K63R, D65H, K66H, V67R. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = -29.0966, G6.0 - G7.4 = 4.1500; mouse G7.4 = 8.3285, G6.0 - G7.4 = 5.4732. G7.4 changes versus the parent are -6.4302 and -1.2649, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 49.88 nM; mouse 59.55 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
AATSVSIALTAPVGAYLVSAYVGGQLVAAEPLRTDAAGDVSKTLSVAALASGDAVSLHVWDVRSHHRVAS
EAV
```

5.12 P02-d070 - bound structures

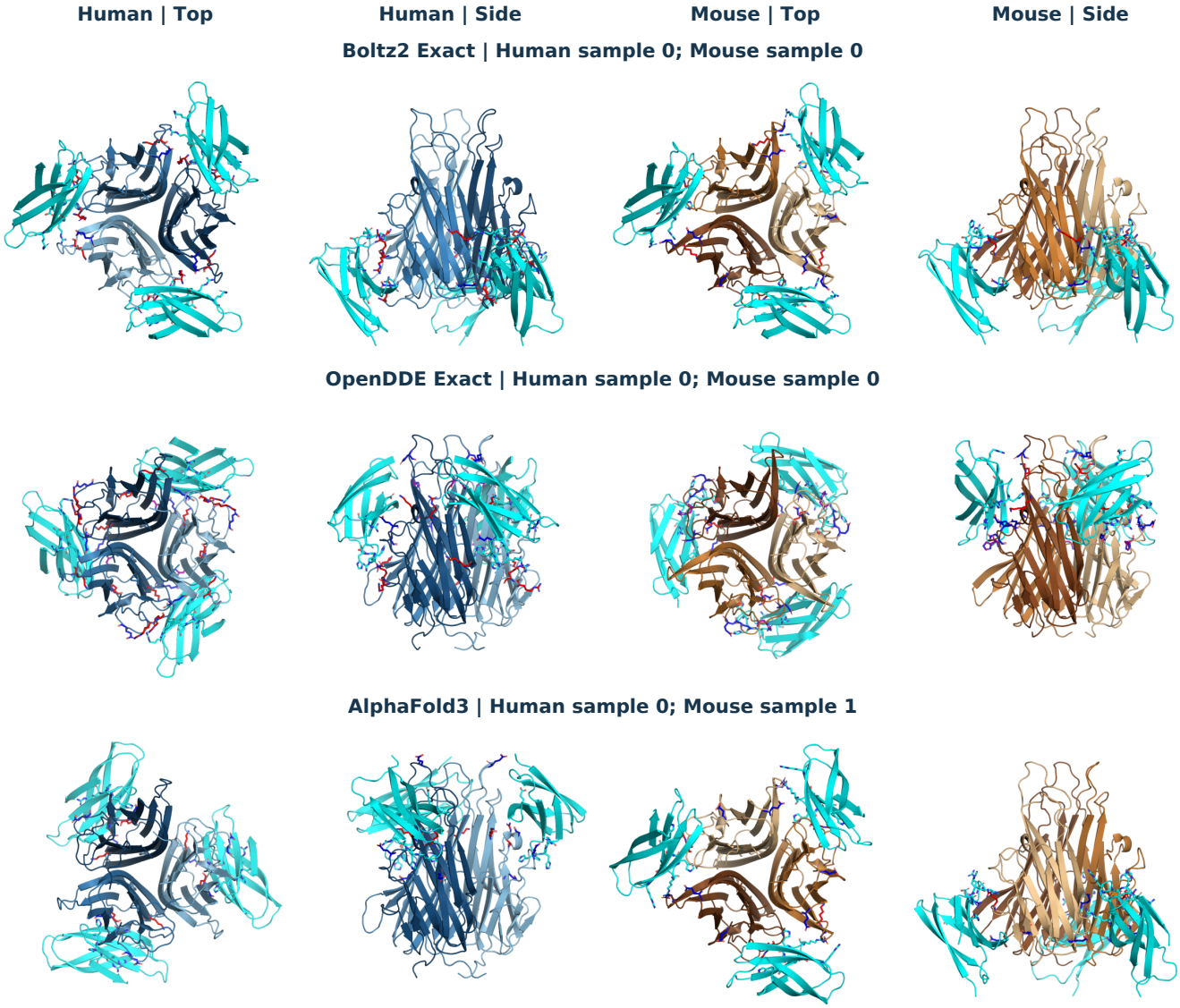


Figure 13. P02-d070: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 4/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.693	0.1222	0.004672
Boltz2 / mouse / 0	0.687	0.1861	0.005234
OpenDDE / human / 0	0.018	0.1547	0.002648
OpenDDE / mouse / 0	0.514	-1.598	0.002472
AF3 / human / 0	0.000	-0.03634	0.004125
AF3 / mouse / 1	0.000	0.2166	0.00387

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 65, 66.

5.13 F25

Rank 13 | GS fusion | 180 residues | Evidence-tier retained

Candidate: F25. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, RP11-d064 and RP03-d017, were joined in that N-to-C order through an exact 19-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - RP11-d064: creation history

RFD3 source scaffold. The source scaffold for RP11-d064 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 80-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 2; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP11-d064. The recorded redesign seed is 21361043, accepted proposal attempt 69, and temperature 0.1, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -655.969112 in the parent to -663.136294; these are model objective values.

Mutable one-based positions. 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 54, 55, 56, 57, 58, 59, 64. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. L35M, T36V, A40G, I41V, A42D, S43H, G56K, S57H, S59A. The 9 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 19.0508, G6.0 - G7.4 = 3.8665; mouse G7.4 = -13.6937, G6.0 - G7.4 = 4.5458. G7.4 changes versus the parent are -10.4492 and -17.1299, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.13 F25 - second domain history

Domain 2 - RP03-d017: creation history

RFD3 source scaffold. The source scaffold for RP03-d017 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d017. The recorded redesign seed is 20561019, accepted proposal attempt 23, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -672.193798; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N. The 8 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 53.9802, G6.0 - G7.4 = 0.7972; mouse G7.4 = 108.5008, G6.0 - G7.4 = 3.3428. G7.4 changes versus the parent are -10.8977 and -11.6783, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified RP11-d064 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 42.05 angstrom and maximum solvent-route length 43.50 angstrom. Domain 1 occupies residues 1-80, the linker 81-99, and domain 2 100-180. Exact linker: GGGGSGGGGSGGGGSGGGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 45.58 nM; mouse 33.23 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
VSTATLPAPGDSTTTPPAASATVTVLSGKVSLTAMVGGAGVDHATLTPGQSHAFPKHAAQMTLVATNVGT
ATATIEYNTTGGGGSGGGGSGGGGSGGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGAT
IAAAAAARATFTVPATPGTYTVTAANGADASSATVTVAAA
```

5.13 F25 - bound structures

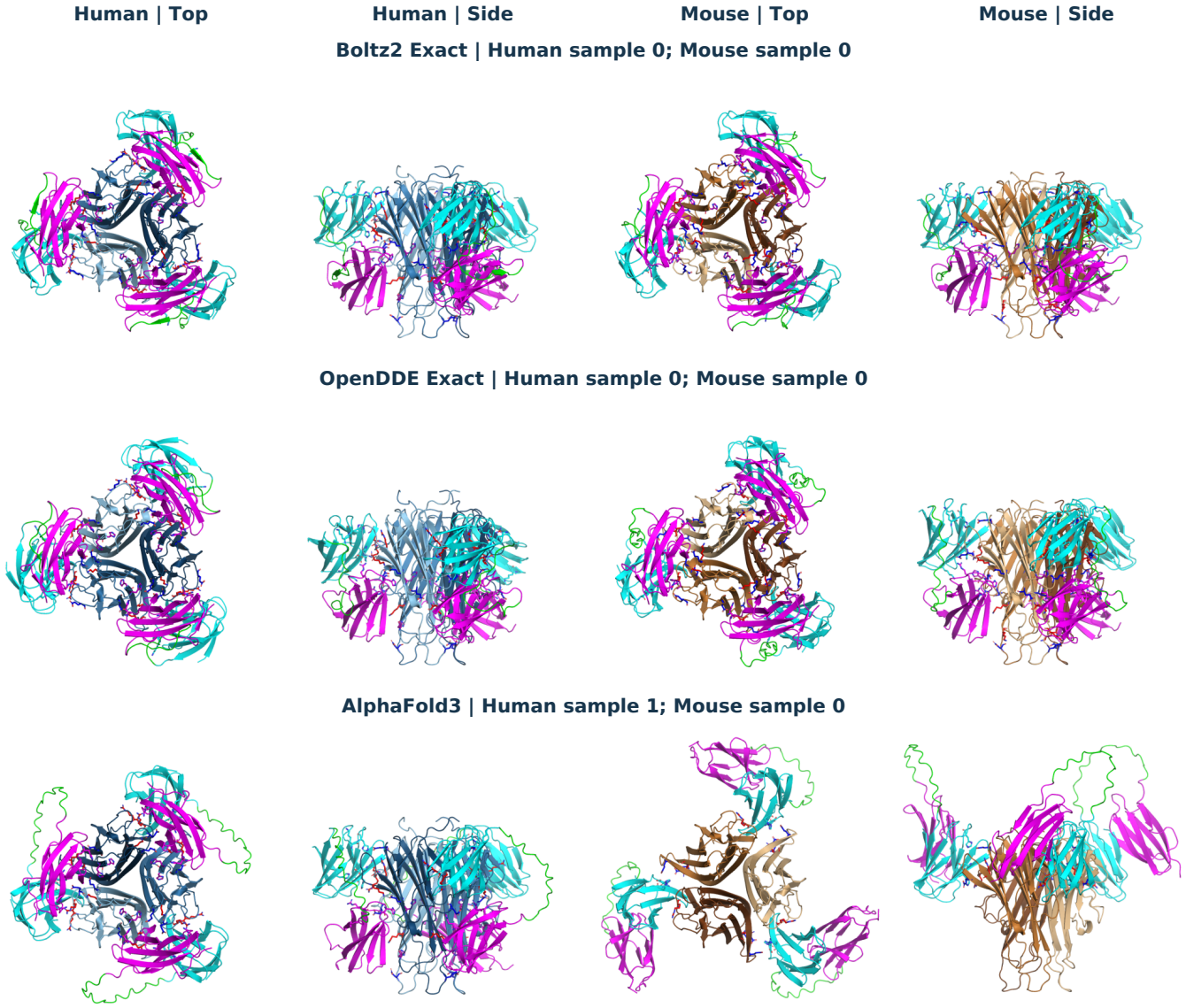


Figure 14. F25: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 3/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.654	-0.7163	-0.01296
Boltz2 / mouse / 0	0.795	-0.6824	-0.01926
OpenDDE / human / 0	0.036	0.008617	1.499e-05
OpenDDE / mouse / 0	0.047	0.01152	0.000178
AF3 / human / 1	0.000	0.4393	0.003987
AF3 / mouse / 0	0.000	0.2479	-0.01544

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 57, 43, 129, 137.

5.14 RP06-d085-V60S

Rank 14 | expert mutant | 83 residues | Evidence-tier retained

Candidate: RP06-d085-V60S. The structures and metrics below refer to its exact selected sequence.

Creation route. Foundry/RFD3 generation and Proton-PottsMPNN redesign produced RP06-d085; the final sequence then received expert-specified V60S substitution(s). The model-guided and expert-specified edits are recorded separately below.

RP06-d085: creation history

RFD3 source scaffold. The source scaffold for RP06-d085 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 83-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 7; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP06-d085. The recorded redesign seed is 20861028, accepted proposal attempt 87, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -715.992329 in the parent to -719.792428; these are model objective values.

Mutable one-based positions. 1, 36, 39, 40, 41, 62, 64, 66, 68, 69, 70, 71, 72, 73, 74, 75, 76, 77, 78, 79. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. S36H, V40R, S41Q, S62T, S64T, V66T, S69G, A71G, S73T, A74N, A75G, D76A, V78T. The 13 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 144.7273, G6.0 - G7.4 = 3.6622; mouse G7.4 = 46.6635, G6.0 - G7.4 = 1.5454. G7.4 changes versus the parent are -10.7404 and -0.2608, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact expert mutation step

Expert-specified V60S substitutions were applied to RP06-d085. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP06-d085-V60S. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 65.39 nM; mouse 65.44 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

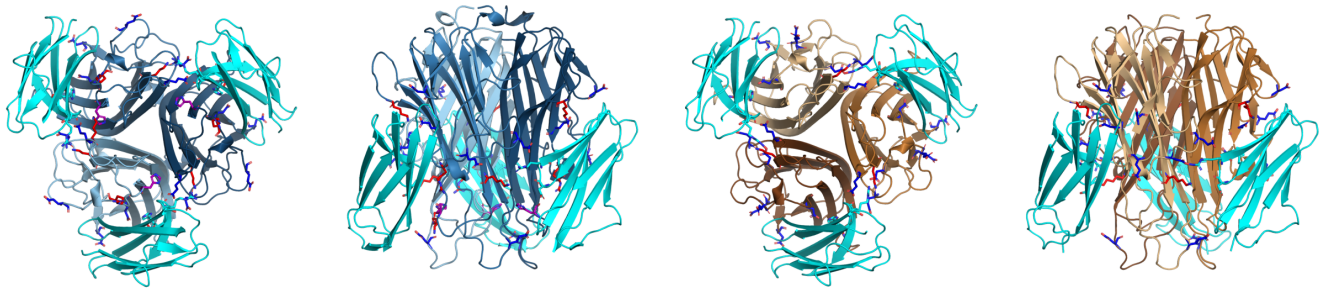
Exact selected sequence (N to C):

```
MAVTLQVAGNVTTVPAGATVTVTCAAAAAATTVTLHADGRQAAATTGSSASVSVTPTAGSHTYTCTATGA
GGTNGASTTVTVT
```

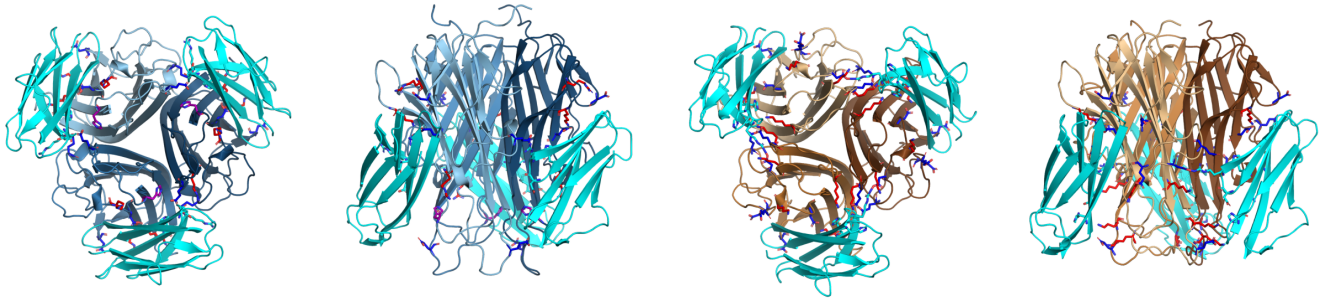
5.14 RP06-d085-V60S - bound structures

Human | Top **Human | Side** **Mouse | Top** **Mouse | Side**

Boltz2 Exact | Human sample 0; Mouse sample 0



OpenDDE Exact | Human sample 0; Mouse sample 0



AlphaFold3 | Human sample 2; Mouse sample 2

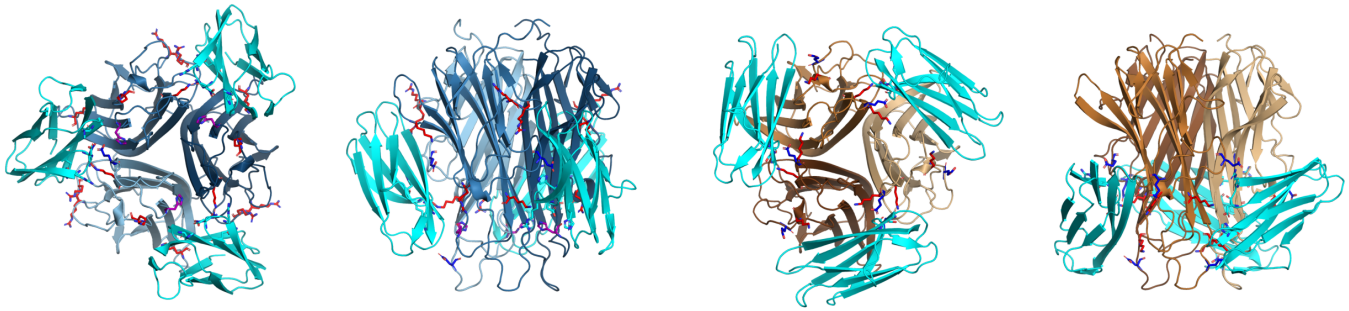


Figure 15. RP06-d085-V60S: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 3/6; ipSAE pass: 3/6; cross-species pose pass: 2/3; intermethod pose pass: 2/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.823	-0.07061	0.0006047
Boltz2 / mouse / 0	0.852	0.01056	-0.008794
OpenDDE / human / 0	0.496	0.00449	1.317e-05
OpenDDE / mouse / 0	0.656	0.006253	-0.006721
AF3 / human / 2	0.000	0.01755	0.0001598
AF3 / mouse / 2	0.000	0.01311	1.959e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 36, 38.

5.15 F11

Rank 15 | GS fusion | 163 residues | Evidence-tier retained

Candidate: F11. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P37-d022 and RP03-d081, were joined in that N-to-C order through an exact 22-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P37-d022: creation history

RFD3 source scaffold. The source scaffold for P37-d022 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 60-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 8; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P37-d022. The recorded redesign seed is 3700111, accepted proposal attempt 22, and temperature 0.2, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -405.867972 in the parent to -413.550655; these are model objective values.

Mutable one-based positions. 3, 12, 14, 15, 16, 39, 40, 41, 42, 43, 44, 45, 46, 47, 48, 49, 52, 53, 54, 57. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. E3H, A12E, A16H, A39P, I42V, V45H, Q46H, T47A, Q48P, I53V. The 10 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 13.3211, G6.0 - G7.4 = 5.7922; mouse G7.4 = -23.4156, G6.0 - G7.4 = 2.5233. G7.4 changes versus the parent are -7.4522 and -2.3585, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.15 F11 - second domain history

Domain 2 - RP03-d081: creation history

RFD3 source scaffold. The source scaffold for RP03-d081 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d081. The recorded redesign seed is 20561019, accepted proposal attempt 111, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -671.910564; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, T21R, S28T, A49H, S51V, Q67S. The 6 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 47.7405, G6.0 - G7.4 = 3.4747; mouse G7.4 = 112.0606, G6.0 - G7.4 = 0.1652. G7.4 changes versus the parent are -17.1374 and -8.1185, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P37-d022 → linker → RP03-d081. The maximum C-to-N distance over the paired geometry was 48.98 angstrom and maximum solvent-route length 52.12 angstrom. Domain 1 occupies residues 1-60, the linker 61-82, and domain 2 83-163. Exact linker: GGGGSGGGGSGGGGSGGGGSGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 37.18 nM; mouse 21.21 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
EAHVTVTVAVPEDATHADILDLALPAAAQLTASDKPVEPVAVAFHHAPDGKLVAKVTFRVGGGGSGGGGS
GGGGSGGGGSGSAAGPAVSAGASTTVTAGARLRVSASNATGAVTTTLSKDGATIAAAAATHAVFTVPATP
GTYTVTAASGADASSATVTVAAA
```

5.15 F11 - bound structures

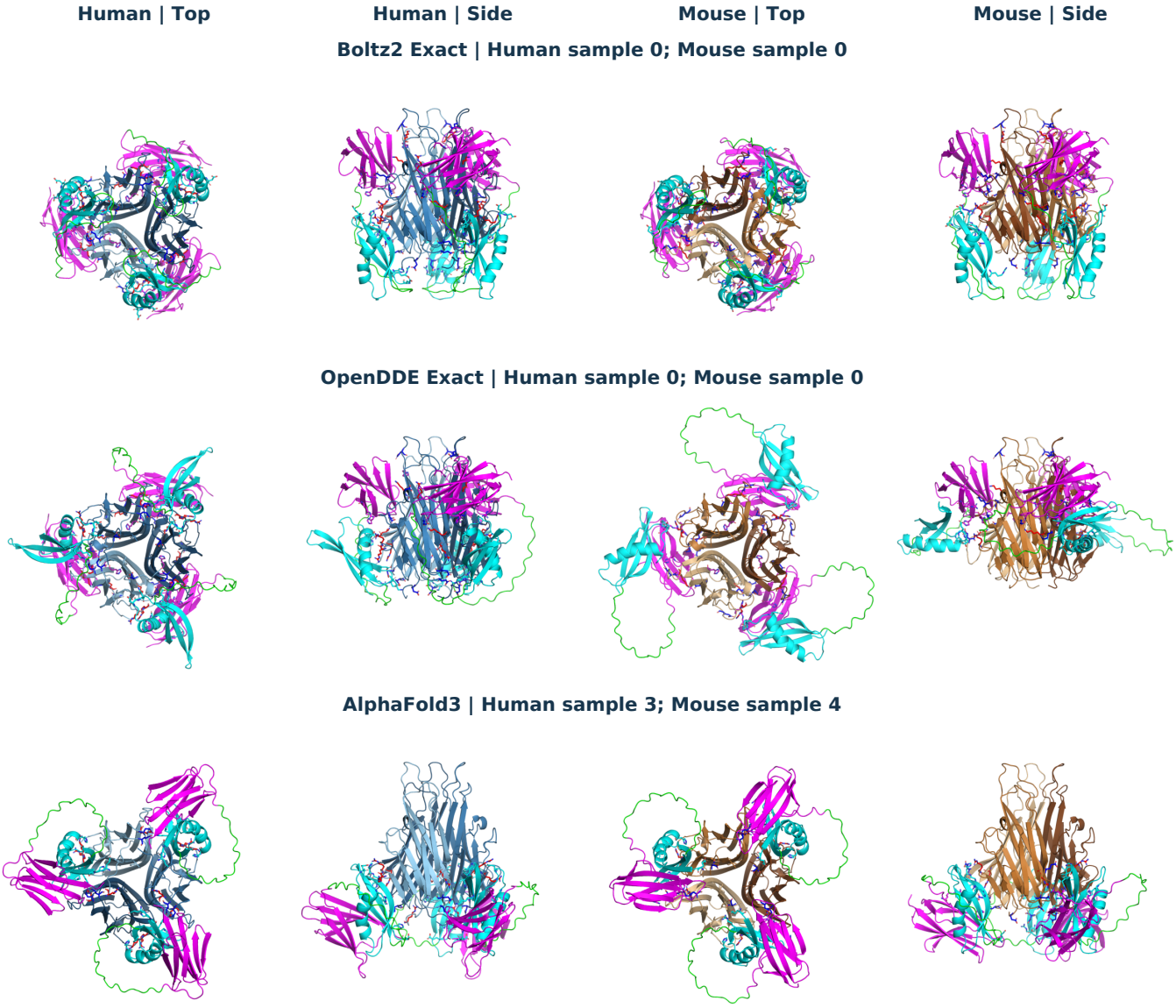


Figure 16. F11: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 3/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 0/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.753	0.03785	0.0001427
Boltz2 / mouse / 0	0.710	0.03676	0.0005054
OpenDDE / human / 0	0.052	0.1366	0.001704
OpenDDE / mouse / 0	0.012	0.8863	-0.02054
AF3 / human / 3	0.128	0.5519	-0.008957
AF3 / mouse / 4	0.000	-3.231	0.001497

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 46, 1, 131, 120.

5.16 F32

Rank 16 | GS fusion | 179 residues | Evidence-tier retained

Candidate: F32. The structures and metrics below refer to its exact selected sequence.

Creation route. Two independently redesigned domains, P30-d032 and RP03-d017, were joined in that N-to-C order through an exact 18-residue GS linker. Both domain histories are reconstructed before the fusion step.

Domain 1 - P30-d032: creation history

RFD3 source scaffold. The source scaffold for P30-d032 used top epitope / mouse conditioning on the complete target trimer: B:LYS112, A:ASP71, A:ASP104, B:GLU110, B:GLU107. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 80-residue parent.

Exact hotspot atom selectors. B112: CB,CG,CD,CE,NZ; A71: CB,CG,OD1,OD2; A104: CB,CG,OD1,OD2; B110: CB,CG,CD,OE1,OE2; B107: CB,CG,CD,OE1,OE2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 1; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P30-d032. The recorded redesign seed is 3000090, accepted proposal attempt 33, and temperature 0.1, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -655.969112 in the parent to -663.172171; these are model objective values.

Mutable one-based positions. 32, 33, 34, 35, 36, 37, 38, 39, 40, 41, 42, 43, 44, 54, 55, 56, 57, 58, 59, 64. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. L35M, T36V, A40G, I41V, A42D, S43H, G56P, S57H. The 8 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 16.4747, G6.0 - G7.4 = 3.8044; mouse G7.4 = -12.5324, G6.0 - G7.4 = 4.4711. G7.4 changes versus the parent are -13.0253 and -15.9687, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

5.16 F32 - second domain history

Domain 2 - RP03-d017: creation history

RFD3 source scaffold. The source scaffold for RP03-d017 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d017. The recorded redesign seed is 20561019, accepted proposal attempt 23, and temperature 0.4, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -672.193798; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, S28T, A30H, T32L, T48A, A49R, S51T, Q67N. The 8 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 53.9802, G6.0 - G7.4 = 0.7972; mouse G7.4 = 108.5008, G6.0 - G7.4 = 3.3428. G7.4 changes versus the parent are -10.8977 and -11.6783, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact fusion construction and subsequent validation

The parent-pair geometry screen specified P30-d032 → linker → RP03-d017. The maximum C-to-N distance over the paired geometry was 41.23 angstrom and maximum solvent-route length 42.98 angstrom. Domain 1 occupies residues 1-80, the linker 81-98, and domain 2 99-179. Exact linker: GGGGSGGGGSGGGGSGGS. No additional substitutions were introduced by concatenation.

The native input specified three separate identical intact fusion copies bound to each TNF trimer. Boltz2 Exact refolding used five diffusion samples per species, seed 0, 200 steps, three recycles, no templates, and native target disulfides. Copy/domain ipSAE, receptor contacts, interdomain/intercopy clashes, and linker geometry were evaluated in the fusion workflow. Later intact-fusion pH/KD assessment followed the initial pending status; inherited parent pH preference was not used as a substitute for this reassessment.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 37.37 nM; mouse 23.37 nM. One intact fusion D was extracted from an earlier three-copy complex, omitting E/F. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
VSTATLPAPGDSTTTPPAASATVTVLSGKVSLTAMVGGAGVDHATLTPGQSHAFPPHASQMTLVATNVGT
ATATIEYNTTGGGGSGGGGSGGGGSGGSAAGPAVSAGASTTVTAGARLTVSASNATGHVLTTLSKDGATI
AAAAAARATFTVPATPGTYTVTAANGADASSATVTVAAA
```

5.16 F32 - bound structures

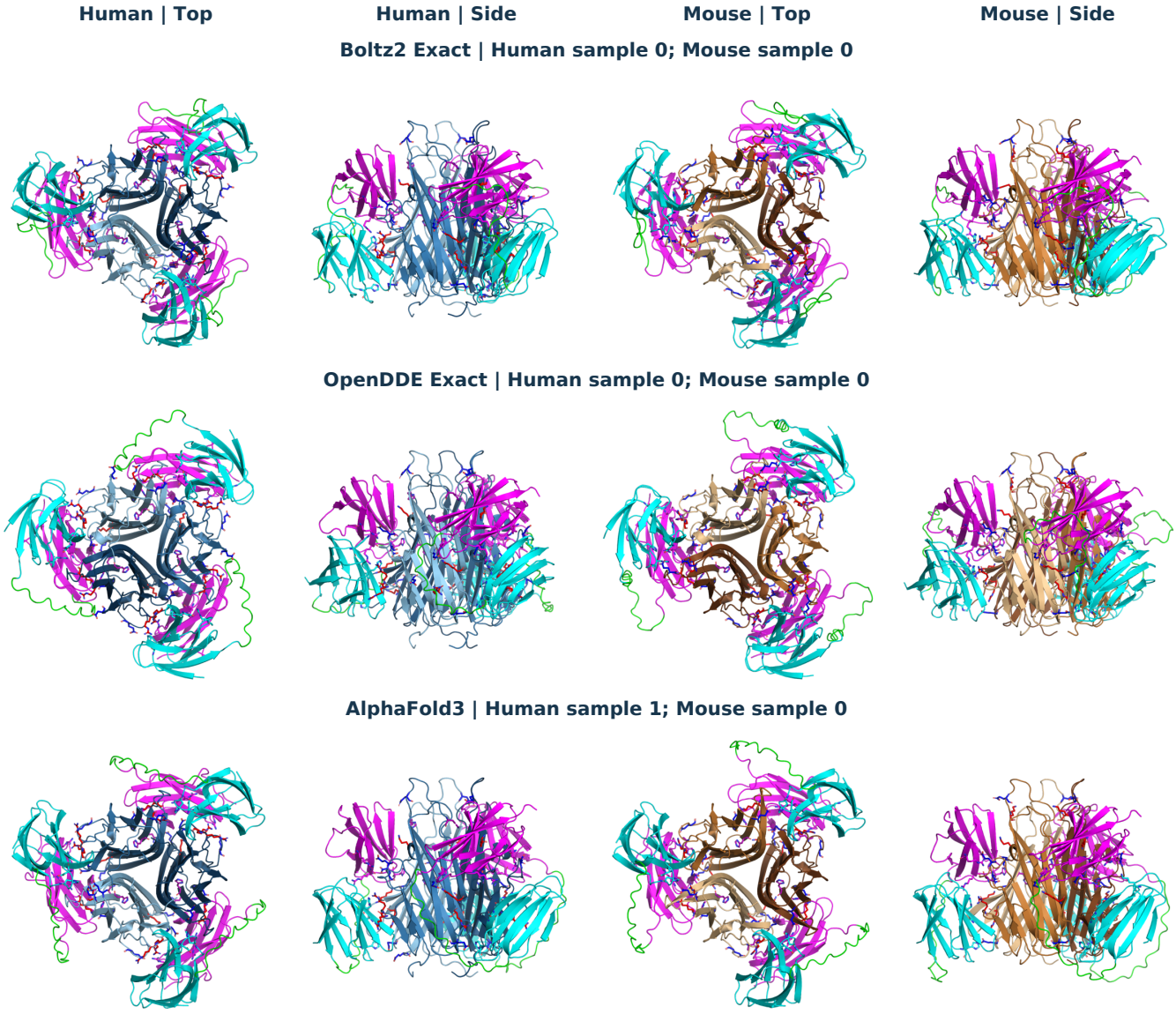


Figure 17. F32: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Within each design copy D/E/F, domain 1 is cyan, the GS linker is green, and domain 2 is magenta.

Current review. Robust-positive pH: 2/6; ipSAE pass: 2/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: ipsae, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.692	0.004646	-0.0191
Boltz2 / mouse / 0	0.826	-0.161	-0.02684
OpenDDE / human / 0	0.031	0.003215	0.0002963
OpenDDE / mouse / 0	0.066	0.03739	0.000438
AF3 / human / 1	0.010	0.01382	-0.008761
AF3 / mouse / 0	0.352	0.1432	-0.05533

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 57, 43, 128, 136.

5.17 RP03-d050-A40N_A69Q

Rank 17 | expert mutant | 81 residues | Evidence-tier retained

Candidate: RP03-d050-A40N_A69Q. The structures and metrics below refer to its exact selected sequence.

Creation route. Foundry/RFD3 generation and Proton-PottsMPNN redesign produced RP03-d050; the final sequence then received expert-specified A40N;A69Q substitution(s). The model-guided and expert-specified edits are recorded separately below.

RP03-d050: creation history

RFD3 source scaffold. The source scaffold for RP03-d050 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d050. The recorded redesign seed is 20561019, accepted proposal attempt 73, and temperature 0.1, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -670.558797; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T21S, S28T, A49H, S51H, Q67T. The 5 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 54.8035, G6.0 - G7.4 = 2.4292; mouse G7.4 = 115.1998, G6.0 - G7.4 = 0.4890. G7.4 changes versus the parent are -10.0744 and -4.9792, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact expert mutation step

Expert-specified A40N;A69Q substitutions were applied to RP03-d050. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP03-d050-A40N_A69Q. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 62.17 nM; mouse 73.52 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
AAGPAVSAGASTTVTAGATLSVSASNATGAVTTTLSKDGNTIAAAAATHAHFTVPATPGTYTVTAATGQD
ASSATVTVAAA
```

5.17 RP03-d050-A40N_A69Q - bound structures

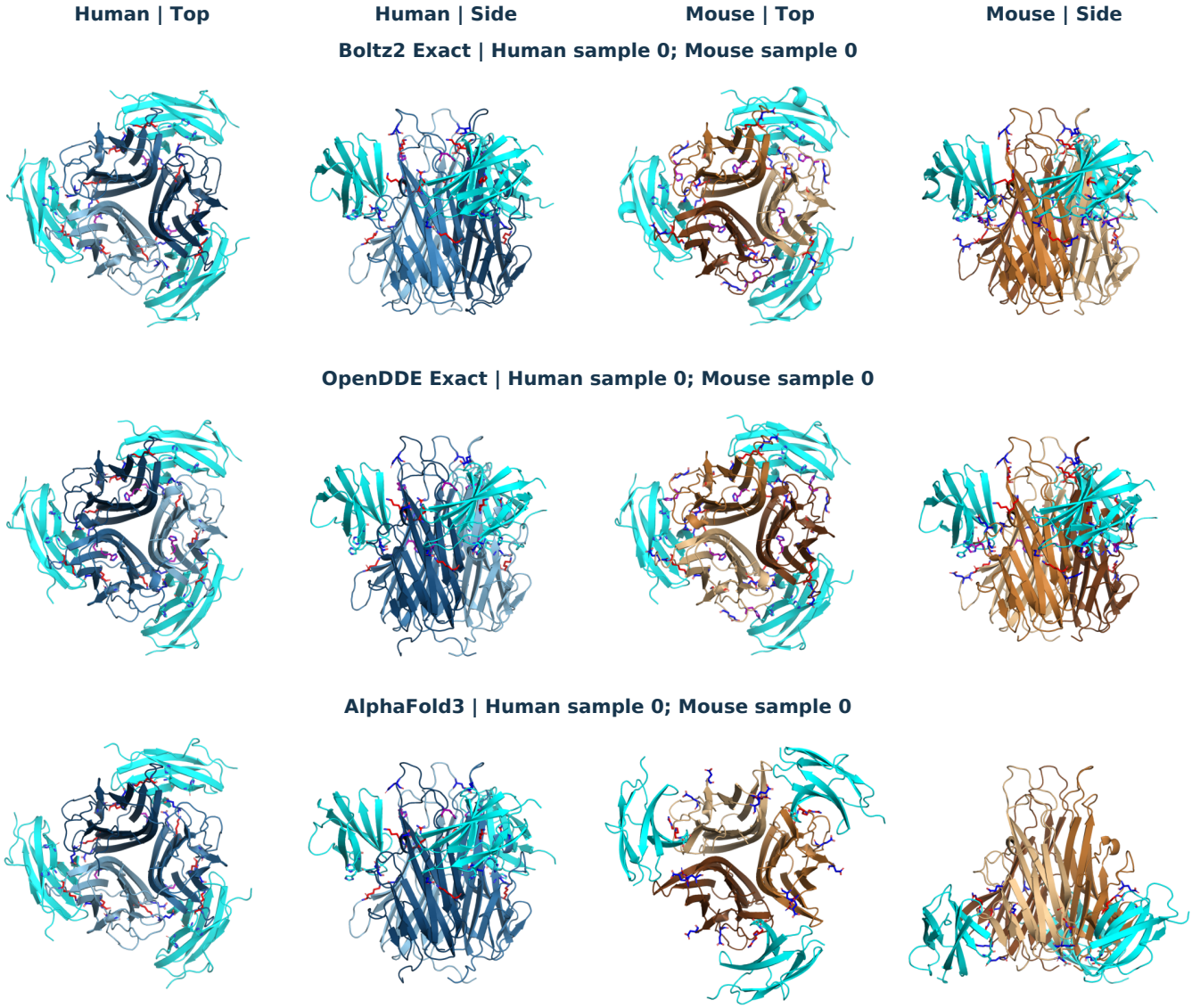


Figure 18. RP03-d050-A40N_A69Q: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 2/6; ipSAE pass: 3/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.839	-0.3741	0.001286
Boltz2 / mouse / 0	0.867	0.04834	0.0009367
OpenDDE / human / 0	0.520	0.06896	-0.02201
OpenDDE / mouse / 0	0.424	0.08944	0.002928
AF3 / human / 0	0.000	0.08843	-0.03171
AF3 / mouse / 0	0.257	0.3403	-0.0408

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 51, 49.

5.18 RP03-d043-A40N_A69Q

Rank 18 | expert mutant | 81 residues | Evidence-tier retained

Candidate: RP03-d043-A40N_A69Q. The structures and metrics below refer to its exact selected sequence.

Creation route. Foundry/RFD3 generation and Proton-PottsMPNN redesign produced RP03-d043; the final sequence then received expert-specified A40N;A69Q substitution(s). The model-guided and expert-specified edits are recorded separately below.

RP03-d043: creation history

RFD3 source scaffold. The source scaffold for RP03-d043 used top epitope / human conditioning on the complete target trimer: C:ARG103, A:GLU104, A:GLU110, A:GLU107, A:ALA109. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 81-residue parent.

Exact hotspot atom selectors. C103: CB,CG,CD,NE,CZ,NH1,NH2; A104: CB,CG,CD,OE1,OE2; A110: CB,CG,CD,OE1,OE2; A107: CB,CG,CD,OE1,OE2; A109: CB.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the later top-epitope selection using mean ipSAE multiplied by binder-fold preservation. Its within-cohort parent rank was 4; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced RP03-d043. The recorded redesign seed is 20561019, accepted proposal attempt 64, and temperature 0.05, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -666.789439 in the parent to -669.254062; these are model objective values.

Mutable one-based positions. 19, 21, 26, 27, 28, 29, 30, 31, 32, 45, 46, 47, 48, 49, 50, 51, 52, 53, 67, 68. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. T19R, A27H, T32L, T48S, A49R, S51T. The 6 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 57.6741, G6.0 - G7.4 = 0.8766; mouse G7.4 = 104.6333, G6.0 - G7.4 = 0.1625. G7.4 changes versus the parent are -7.2038 and -15.5458, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Exact expert mutation step

Expert-specified A40N;A69Q substitutions were applied to RP03-d043. All other residues and the sequence length were unchanged, checked against the parent. The resulting candidate is RP03-d043-A40N_A69Q. Independent human/mouse refolding and downstream scoring assessed the changed sequence; parental pH and affinity values were not transferred.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 79.22 nM; mouse 74.35 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
AAGPAVSAGASTTVTAGARLTVSASNHSGAVLTTLSKDGNTIAAAAASRATFTVPATPGTYTVTAAQGQD
ASSATVTVAAA
```

5.18 RP03-d043-A40N_A69Q - bound structures

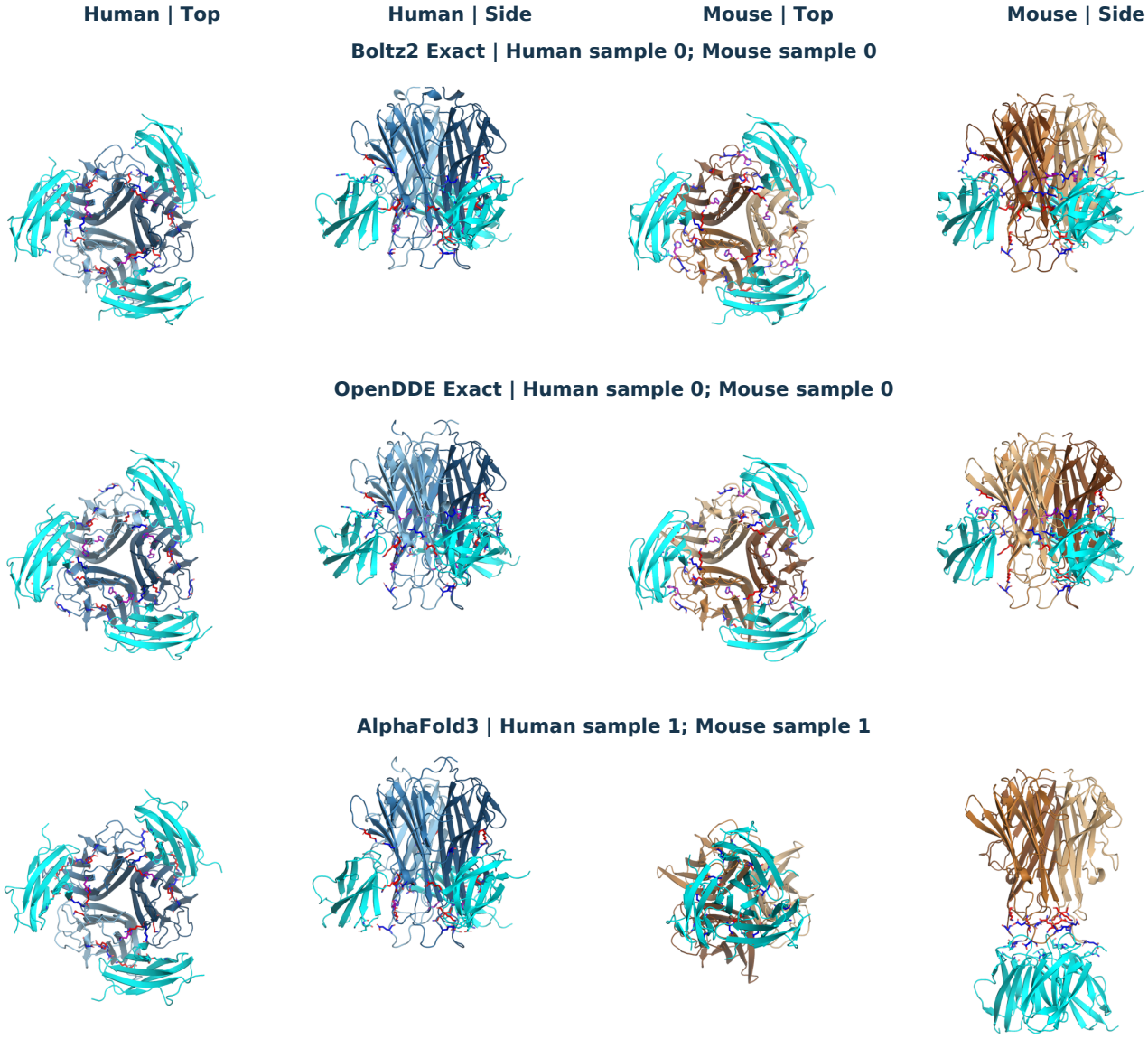


Figure 19. RP03-d043-A40N_A69Q: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 2/6; ipSAE pass: 2/6; cross-species pose pass: 2/3; intermethod pose pass: 4/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.798	0.01687	-0.007585
Boltz2 / mouse / 0	0.877	0.02599	0.002199
OpenDDE / human / 0	0.341	0.003953	-0.01091
OpenDDE / mouse / 0	0.429	0.1631	-0.02576
AF3 / human / 1	0.293	0.06885	-0.005195
AF3 / mouse / 1	0.000	0.002861	0.001047

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 27, 70.

5.19 P16-d085

Rank 19 | Proton-Potts redesign | 75 residues | Evidence-tier retained

Candidate: P16-d085. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P16-d085: creation history

RFD3 source scaffold. The source scaffold for P16-d085 used side epitope / mouse conditioning on the complete target trimer: A:LYS65, A:LYS112, A:GLU146, C:HIS78, C:LYS90, C:LYS140. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 75-residue parent.

Exact hotspot atom selectors. A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C78: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C140: CB,CG,CD,CE,NZ.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 7; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P16-d085. The recorded redesign seed is 1600048, accepted proposal attempt 88, and temperature 0.05, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -533.628685 in the parent to -543.002282; these are model objective values.

Mutable one-based positions. 27, 28, 29, 31, 32, 33, 34, 37, 38, 57, 58, 59, 63, 64, 65, 66, 67, 68, 69, 72. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. N27D, T29V, N32G, K33R, A34T, D37H, A59T, Q64K, T67H, F68H, K69R, T72H. The 12 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 89.8002, G6.0 - G7.4 = 6.4521; mouse G7.4 = 54.9628, G6.0 - G7.4 = 3.3822. G7.4 changes versus the parent are -13.7211 and -15.2718, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 29.04 nM; mouse 24.05 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
AVSITLPAASASPPIVITGDSESTVADVVKDGRTVAHNLQSDAAGTLSITLAPGAYLITAADLKSGHHRT
AHVTV
```

5.19 P16-d085 - bound structures

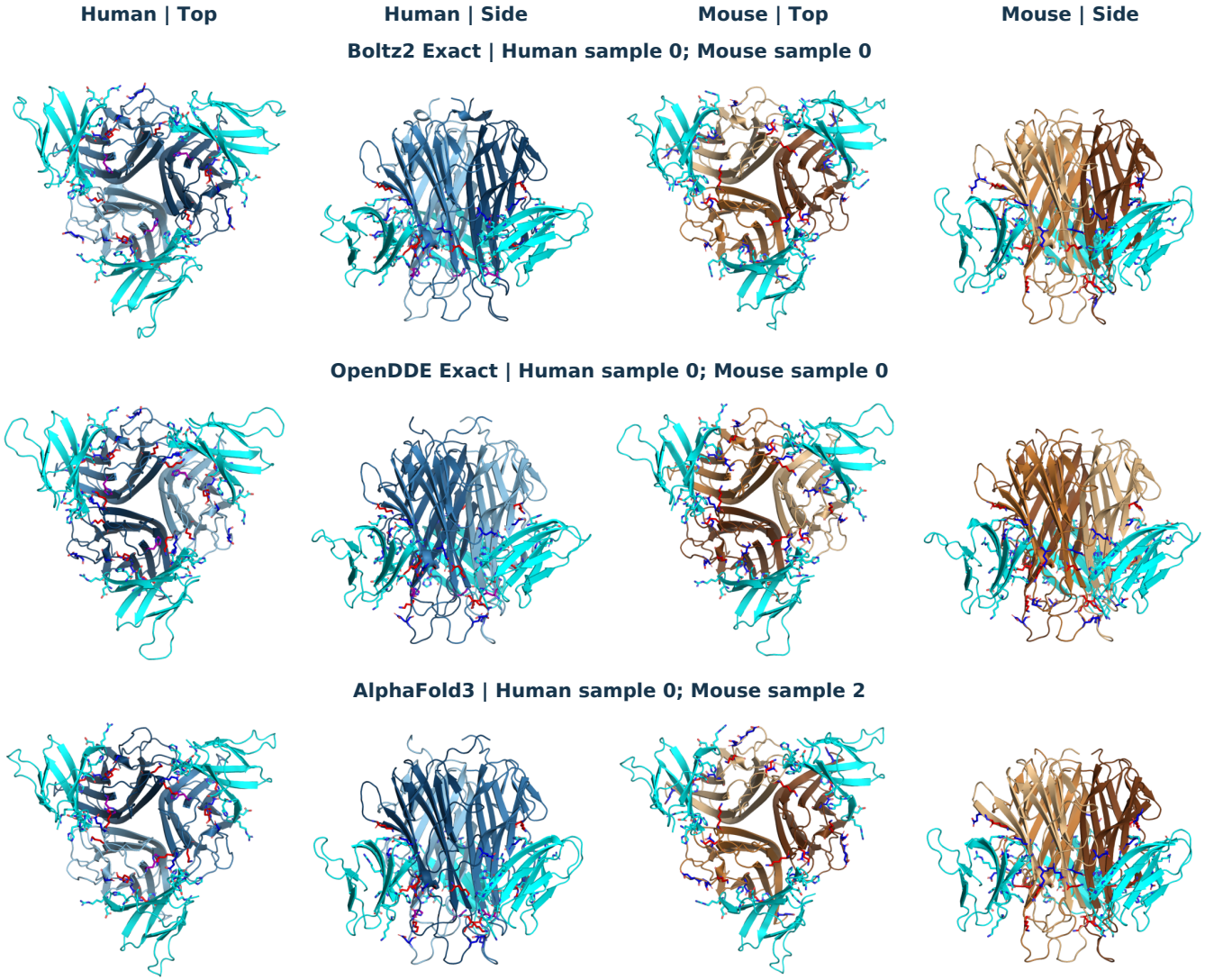


Figure 20. P16-d085: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 1/6; ipSAE pass: 6/6; cross-species pose pass: 3/3; intermethod pose pass: 6/6. Failed checks: pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.877	-4.676	0.002135
Boltz2 / mouse / 0	0.895	0.001136	-0.003174
OpenDDE / human / 0	0.566	-2.644	0.00237
OpenDDE / mouse / 0	0.678	0.3282	0.01384
AF3 / human / 0	0.713	-2.93	0.004733
AF3 / mouse / 2	0.760	-0.5447	1.586e-05

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 72, 27.

5.20 P00-d000

Rank 20 | Proton-Potts redesign | 63 residues | Evidence-tier retained

Candidate: P00-d000. The structures and metrics below refer to its exact selected sequence.

Creation route. The selected molecule is the intact Proton-PottsMPNN descendant of its hotspot-conditioned RFD3 parent. Its recorded sequence-changing steps are source RFD3 generation and the explicit Proton-PottsMPNN substitutions below.

P00-d000: creation history

RFD3 source scaffold. The source scaffold for P00-d000 used side epitope / human conditioning on the complete target trimer: A:LYS65, A:LYS112, A:GLU146, C:HIS73, C:LYS90, C:ARG138. The retained specification records the species contig and fixed-target settings in section 2.1, hotspot-based orientation, and one separate 60-100-residue design chain. Coordinate-derived sequence export supplied the exact 63-residue parent.

Exact hotspot atom selectors. A65: CB,CG,CD,CE,NZ; A112: CB,CG,CD,CE,NZ; A146: CB,CG,CD,OE1,OE2; C73: CB,CG,ND1,CD2,CE1,NE2; C90: CB,CG,CD,CE,NZ; C138: CB,CG,CD,NE,CZ,NH1,NH2.

Paired parent refolding and admission. Separate human and mouse Boltz2 backbones supported the parent selection. The scaffold entered the initial four-cohort selection using mean ipSAE multiplied by human/mouse binder-pose agreement. Its within-cohort parent rank was 1; this is distinct from the current candidate rank.

Proton-PottsMPNN proposal and optimization. Fixed paired backbones produced P00-d000. The recorded redesign seed is 0, accepted proposal attempt 0, and temperature 0.05, with 15 optimization sweeps under the shared two-pH objective and acceptance rules. The composite objective changed from -515.213537 in the parent to -516.811279; these are model objective values.

Mutable one-based positions. 7, 8, 10, 11, 12, 13, 14, 27, 28, 29, 30, 31, 32, 33, 35, 36, 37, 39, 40, 41. All remaining residues, chain length, and native Cys were fixed.

Complete substitutions from the RFD3 parent. S10A, A11S, A28E, T30L, L31R, A32V, A40S. The 7 differences were checked directly against the frozen parent and descendant sequences.

Generation-time pH guard and sequence handoff. Human G7.4 = 112.9613, G6.0 - G7.4 = 1.7788; mouse G7.4 = 153.6164, G6.0 - G7.4 = 0.9637. G7.4 changes versus the parent are -24.9185 and -23.9977, respectively. The exact accepted sequence was exported to native FASTA and paired Boltz2 refolding after redesign. These intrinsic-pKa generation values are distinct from later pose-specific PROPKA assessments.

Downstream refolding and scoring. Paired Boltz2 structures were followed by native-pose pH assessment and separate DeltaForge affinity prediction. The later six-model comparison below assesses the exact sequence across both species and all three prediction methods.

Archived DeltaForge KD: human 17.26 nM; mouse 13.07 nM. The scored object was an earlier A/B/C + single candidate D pose. No new affinity calculation for this panel is available.

Exact selected sequence (N to C):

```
SAKAVSAALASIGLNAGVTSTISTAGSEALRVAIRAAAASHGLQAVVVEGADGASAIRLANAA
```

5.20 P00-d000 - bound structures

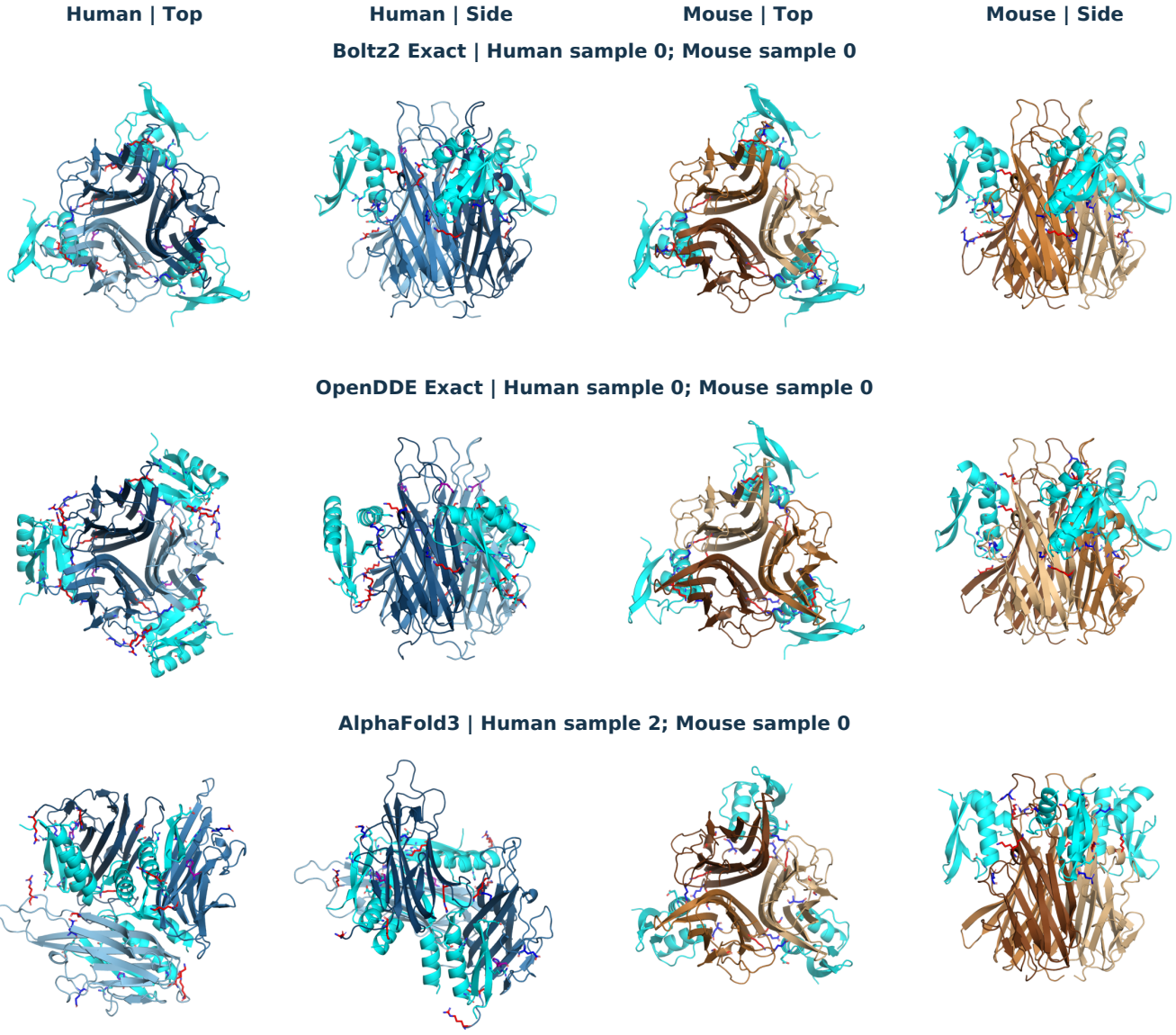


Figure 21. P00-d000: adjacent top and side views for human (blue target) and mouse (copper target). Side views rotate the camera 90 degrees, retaining a common alignment and scale. Single-domain design copies D/E/F are cyan.

Current review. Robust-positive pH: 1/6; ipSAE pass: 3/6; cross-species pose pass: 1/3; intermethod pose pass: 1/6. Failed checks: ipsae, cross pose, intermethod, pH.

Model / species / sample	Min-copy ipSAE	Robust Potts	Robust MPNN
Boltz2 / human / 0	0.900	-2.461	0.009805
Boltz2 / mouse / 0	0.904	-2.528	0.002463
OpenDDE / human / 0	0.000	0.4075	0.007091
OpenDDE / mouse / 0	0.794	-2.543	0.0007416
AF3 / human / 2	0.014	-1.78	0.001561
AF3 / mouse / 0	0.000	-0.199	0.02502

Potts and MPNN columns are separate score units (section 2.4), each minimized over D/E/F and pKa shifts -1/0/+1. Positive favors pH 7.4. Exact titration-center sequence positions: 28, 41.

6 Interpreting the computational evidence

The selected set is a computational review portfolio. The six species/method outcomes share sequences, structures, scoring models, and assumptions; they are not six independent experimental observations. Positive contrasts at the selected titration centers and pKa scenarios support prioritization for experimental testing. The model-specific metrics and structural disagreements remain visible in each candidate section.

Both top and side views are provided for all three prediction methods against both human and mouse TNF-alpha. Each model contains three separate copies of the selected protein bound to the target trimer. Saved DeltaForge affinities concern earlier single-binder poses, while the current structural and pH comparison uses the displayed three-copy complexes. These distinct scopes are retained throughout.

6.1 Method references

References below retain the source bibliography identifiers from the supplied methods. Native records, rather than software names alone, determine which design steps are attributed to each candidate.

3. Dauparas et al. (2022). Robust deep learning-based protein sequence design using ProteinMPNN. Science 378, 49-56. Source

4. Birnbaum and Keating (2026). Beyond native sequence recovery: Improved modeling of the sequence-energy landscape of protein structures. PNAS 123, e2535494123. Source

6. Abramson et al. (2024). Accurate structure prediction of biomolecular interactions with AlphaFold 3. Nature 630, 493-500. Source

7. Corley et al. (2025). Accelerating Biomolecular Modeling with AtomWorks and RF3. bioRxiv. Source

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9. RosettaCommons. Foundry software repository. Accessed 11 October 2026. Source

10. Passaro et al. (2025). Boltz-2: Towards Accurate and Efficient Binding Affinity Prediction. bioRxiv. Source

11. Anthropic. Uplifting biomolecular modeling: inference optimization kits. Accessed 11 October 2026. Source

12. Kalakoti and Wallner (2026). AFsample3: Generating and selecting multiple conformational states with Alphafold3. bioRxiv, version 1 posted 17 January 2026. Source

14. Jacobsen et al. (2026). pH-sensitive binder design with Proton-PottsMPNN. bioRxiv, version 1 posted 1 October 2026. Source

15. Watson, A. (2026). Single-Pass Discrete Diffusion Predicts High-Affinity Peptide Binders at >1,000 Sequences per Second across 150 Receptor Targets. bioRxiv, version 1 posted 17 March 2026. Source

16. Dunbrack (2025). Res ipSAE loquuntur: What's wrong with AlphaFold's ipTM score and how to fix it. bioRxiv. Source

17. Zhang et al. (2022). US-align: universal structure alignments of proteins, nucleic acids, and macromolecular complexes. Nature Methods 19, 1109-1115. Source